

9.46.6 Quadratic Equations: GATE2013 EE: GA-8



The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is ¹

- A. $(-\infty, 0)$ B. $(0, 1)$ C. $(1, \infty)$ D. $(0, \infty)$ ²

gate2013-ee quantitative-aptitude quadratic-equations

Answer key

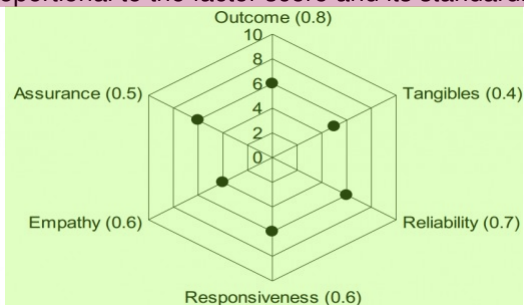
9.47

Radar Chart (1)

9.47.1 Radar Chart: GATE2011 GG: GA-9



The quality of services delivered by a company consists of six factors as shown below in the radar diagram. The dots in the figure indicate the score for each factor on a scale of 0 to 10. The standardized coefficient for each factor is given in the parentheses. The contribution of each factor to the overall service quality is directly proportional to the factor score and its standardized coefficient.



The lowest contribution among all the above factors to the overall quality of services delivered by the company is ⁵

- A. 10% B. 20% C. 24% D. 40% ⁶

gate2011-gg difficult quantitative-aptitude data-interpretation radar-chart ⁷

Answer key ⁸

9.48

Ratio Proportion (9)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (3Q) ⁹

9.48.1 Ratio Proportion: GATE CSE 2016 | Set 1 | GA | Question: 10



In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

- A. 40.00 B. 46.02 C. 60.01 D. 92.02 ¹²

gatecse-2016-set1 quantitative-aptitude ratio-proportion normal ¹³

Answer key

9.48.2 Ratio Proportion: GATE CSE 2018 | GA | Question: 8



In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

- A. 2:3 B. 1:1 C. 3:2 D. 2:1

gatecse-2018 quantitative-aptitude ratio-proportion two-marks

Answer key

9.48.3 Ratio Proportion: GATE CSE 2021 | Set 1 | GA | Question: 1



The ratio of boys to girls in a class is 7 to 3.

1

Among the options below, an acceptable value for the total number of students in the class is:

- A. 21 B. 37 C. 50 D. 73 ²

gatecse-2021-set1 quantitative-aptitude ratio-proportion one-mark ³

Answer key

9.48.4 Ratio Proportion: GATE CSE 2021 | Set 2 | GA | Question: 8



The number of students in three classes is in the ratio 3 : 13 : 6. If 18 students are added to each class, the ratio changes to 15 : 35 : 21.

4

The total number of students in all the three classes in the beginning was: ⁵

- A. 22 B. 66 C. 88 D. 110 ⁶

gatecse-2021-set2 ratio-proportion quantitative-aptitude two-marks ⁷

Answer key

9.48.5 Ratio Proportion: GATE CSE 2024 | Set 1 | GA | Question: 2



If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$ ⁸

- A. depends on xy B. depends only on x and not on y ⁹
C. depends only on y and not on x D. is a constant

gatecse-2024-set1 quantitative-aptitude ratio-proportion one-mark ¹⁰

Answer key

9.48.6 Ratio Proportion: GATE2011 AG: GA-8



Three friends, R , S and T shared toffee from a bowl. R took $\frac{1}{3}^{\text{rd}}$ of the toffees, but returned four to the bowl. S took $\frac{1}{4}^{\text{th}}$ of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

- A. 38 B. 31 C. 48 D. 41 ¹³

general-aptitude quantitative-aptitude gate2011-ag ratio-proportion ¹⁴

Answer key

9.48.7 Ratio Proportion: GATE2011 GG: GA-4



If m students require a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in ¹⁵

- A. 100 days B. $m/100$ days C. $100/m$ days D. m days ¹⁷

gate2011-gg quantitative-aptitude ratio-proportion ¹⁸

Answer key

9.48.8 Ratio Proportion: GATE2013 AE: GA-1



If $3 \leq X \leq 5$ and $8 \leq Y \leq 11$ then which of the following options is TRUE? ²⁰

- A. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$ B. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}\right)$ ²¹
C. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$ D. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}\right)$

gate2013-ae quantitative-aptitude ratio-proportion normal

Answer key ²²

9.48.9 Ratio Proportion: GATE2014 AE: GA-8



The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is $3 : 4 : 5 : 6$. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

gate2014-ae quantitative-aptitude ratio-proportion numerical-answers 2

Answer key 3

9.49 4

Sequence Series (4) 5

Practice Tests: Test 1 (15Q) Test 2 (1Q) 6

9.49.1 Sequence Series: GATE CSE 2012 | Question: 65



Given the sequence of terms, AD CG FK JP, the next term is 7

A. OV B. OW C. PV D. PW 8

gatecse-2012 quantitative-aptitude sequence-series easy 9

Answer key

9.49.2 Sequence Series: GATE CSE 2018 | GA | Question: 5



What is the missing number in the following sequence? 10

2, 12, 60, 240, 720, 1440, _____, 0 11

A. 2880 B. 1440 C. 720 D. 0 12

gatecse-2018 quantitative-aptitude sequence-series easy one-mark 13

Answer key

9.49.3 Sequence Series: GATE CSE 2023 | GA | Question: 3



A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \dots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$.

If $F_6 = 37$, and $F_7 = 60$, then what is F_1 ?

A. 4 B. 5 C. 8 D. 9 15

gatecse-2023 quantitative-aptitude sequence-series one-mark 16

Answer key

9.49.4 Sequence Series: GATE DS&AI 2024 | GA | Question: 4



The sum of the following infinite series is 17

$$2 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{8} + \frac{1}{9} + \frac{1}{16} + \frac{1}{27} + \dots$$

A. $\frac{11}{3}$ B. $\frac{7}{2}$ C. $\frac{13}{4}$ D. $\frac{9}{2}$ 19

gate-ds-ai-2024 quantitative-aptitude sequence-series one-mark 20

Answer key 21

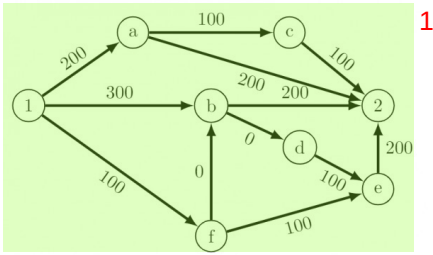
9.50

Shortest Path (1)

9.50.1 Shortest Path: GATE CSE 2020 | GA | Question: 5



There are multiple routes to reach from node 1 to node 2, as shown in the network.



The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

- A. $1 - a - c - 2$ B. $1 - f - b - 2$ C. $1 - b - 2$ D. $1 - f - e - 2$

gatecse-2020 quantitative-aptitude graph-theory shortest-path one-mark

Answer key

9.51

Speed Time Distance (4)

Practice Tests: Test 1 (15Q) Test 2 (11Q)

9.51.1 Speed Time Distance: GATE CSE 2013 | Question: 64

A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

- A. 36 B. 30 C. 24 D. 18

gatecse-2013 quantitative-aptitude easy speed-time-distance

Answer key

9.51.2 Speed Time Distance: GATE CSE 2019 | GA | Question: 3

Two cars at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

- A. 1 B. 2 C. 3 D. 6

gatecse-2019 general-aptitude quantitative-aptitude speed-time-distance one-mark

Answer key

9.51.3 Speed Time Distance: GATE2013 AE: GA-6

Velocity of an object fired directly in upward direction is given by $V = 80 - 32t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

- A. $\left(1, \frac{3}{2}\right)$ B. $\left(\frac{1}{2}, 1\right)$ C. $\left(\frac{1}{2}, \frac{3}{2}\right)$ D. $(1, 3)$

gate2013-ae quantitative-aptitude speed-time-distance

Answer key

9.51.4 Speed Time Distance: GATE2013 EE: GA-9

A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

- A. 30 B. 36 C. 40 D. 24

gate2013-ee speed-time-distance quantitative-aptitude

9.52

Statistics (3)

Practice Test: Test 1 (11Q) 1

9.52.1 Statistics: GATE CSE 2012 | Question: 64

Which of the following assertions are **CORRECT**? 2

- P: Adding 7 to each entry in a list adds 7 to the mean of the list 3
- Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list
- R: Doubling each entry in a list doubles the mean of the list
- S: Doubling each entry in a list leaves the standard deviation of the list unchanged

A. P, Q B. Q, R C. P, R D. R, S 4

gatecse-2012 quantitative-aptitude statistics normal 5

Answer key

9.52.2 Statistics: GATE CSE 2024 | Set 1 | GA | Question: 3



Consider the following sample of numbers: 7

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

A. 13.5 B. 14 C. 11 D. 18.7 8

gatecse-2024-set1 quantitative-aptitude statistics one-mark 9

Answer key

9.52.3 Statistics: GATE2012 AE: GA-5



The arithmetic mean of five different natural numbers is 12. The largest possible value among the numbers is 15

A. 12 B. 40 C. 50 D. 60 12

gate2012-ae statistics quantitative-aptitude 13

Answer key 14

9.53

System of Equations (1)

9.53.1 System of Equations: GATE2011 GG: GA-6



The number of solutions for the following system of inequalities is 15

- $X_1 \geq 0$
- $X_2 \geq 0$
- $X_1 + X_2 \leq 10$
- $2X_1 + 2X_2 \geq 22$

16

A. 0 B. infinite C. 1 D. 2 17

gate2011-gg quantitative-aptitude system-of-equations 18

Answer key 19

9.54

Tables (1)

9.54.1 Tables: GATE CSE 2025 | Set 2 | GA | Question: 8



Which one of the following options is correct for the given data in the table?

Iteration (i)	0	1	2	3
Input (I)	20	-4	10	15
Output (X)	20	16	26	41
Output (Y)	20	-80	-800	-12000

- A. $X(i) = X(i-1) + I(i); Y(i) = Y(i-1)I(i); i > 0$
 B. $X(i) = X(i-1)I(i); Y(i) = Y(i-1) + I(i); i > 0$
 C. $X(i) = X(i-1)I(i); Y(i) = Y(i-1)I(i); i > 0$
 D. $X(i) = X(i-1) + I(i); Y(i) = Y(i-1)I(i-1); i > 0$

gatecse2025-set2 quantitative-aptitude tables two-marks

Answer key

9.55

Tabular Data (7)

Practice Test: Test 1 (12Q)

9.55.1 Tabular Data: GATE CSE 2014 | Set 1 | GA | Question: 9

In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
Own vehicle	Scooter	30	20
Own vehicle	Both	60	46
Do not own vehicle		20	50

gatecse-2014-set1 quantitative-aptitude normal numerical-answers data-interpretation tabular-data

Answer key

9.55.2 Tabular Data: GATE CSE 2014 | Set 3 | GA | Question: 5

The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

- A. 1.34 B. 1.74 C. 3.02 D. 3.91

gatecse-2014-set3 quantitative-aptitude normal data-interpretation tabular-data

Answer key

9.55.3 Tabular Data: GATE CSE 2015 | Set 1 | GA | Question: 6

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

1

What is the average of the marks obtained by the class in the examination? 2

- A. 2.290 B. 2.970 C. 6.795 D. 8.795 3

gatecse-2015-set1 quantitative-aptitude easy data-interpretation tabular-data 4

Answer key

9.55.4 Tabular Data: GATE CSE 2016 | Set 1 | GA | Question: 06



A shaving set company sells 4 different types of razors- Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter/Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

6

Which product contributes the greatest fraction to the revenue of the company in that year? 7

- A. Elegance B. Executive C. Smooth D. Soft 8

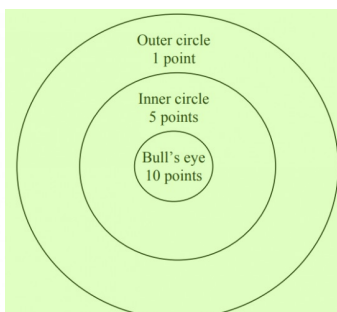
gatecse-2016-set1 quantitative-aptitude data-interpretation easy tabular-data 9

Answer key

9.55.5 Tabular Data: GATE2011 MN: GA-64



Four archers P, Q, R, and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bull's eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.



11

The final scores received by the players during the tournament are listed in the table below. 12

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively ¹ ²

- A. P and S B. Q and R C. Q and Q D. R and Q ³

gate2011-mn data-interpretation quantitative-aptitude tabular-data ⁴

Answer key 

9.55.6 Tabular Data: GATE2013 AE: GA-7

Following table gives data on tourist from different countries visiting India in the year 2011 ⁵



Country	Number of tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011? ⁶ ⁷

- A. USA and Japan B. USA and Australia C. England and France D. Japan and Australia ⁸

gate2013-ae quantitative-aptitude data-interpretation normal tabular-data ⁹

Answer key 

9.55.7 Tabular Data: GATE2013 CE: GA-8

Following table provides figures(in rupees) on annual expenditure of a firm for two years -2010 and 2011. ¹⁰



Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the two categories have registered increase by same percentage? ¹¹ ¹²

- A. Raw material and Salary & wages. B. Salary & wages and Advertising. ¹³
C. Power & fuel and Advertising. D. Raw material and research & Development.

quantitative-aptitude gate2013-ce data-interpretation normal tabular-data

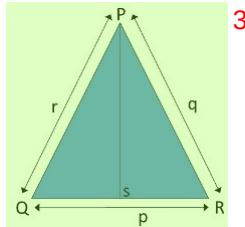
9.56

Triangles (3)

Practice Test: Test 1 (11Q)

9.56.1 Triangles: GATE CSE 2015 | Set 2 | GA | Question: 8

In a triangle PQR , PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS ?



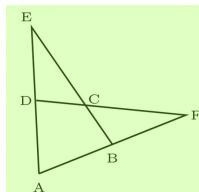
- A. $\left(\frac{q+r}{qr}\right)$ B. $\left(\frac{qr}{q+r}\right)$
 C. $\sqrt{(q^2 + r^2)}$ D. $\left(\frac{(q+r)^2}{qr}\right)$

gatecse-2015-set2 quantitative-aptitude geometry difficult triangles

Answer key

9.56.2 Triangles: GATE CSE 2018 | GA | Question: 9

In the figure below, $\angle DEC + \angle BFC$ is equal to _____



- A. $\angle BCD - \angle BAD$ B. $\angle BAD + \angle BCF$
 C. $\angle BAD + \angle BCD$ D. $\angle CBA + \angle ADC$

gatecse-2018 quantitative-aptitude geometry normal triangles two-marks

Answer key

9.56.3 Triangles: GATE CSE 2022 | GA | Question: 9

The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to the corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

- A. P cannot be placed at a corner B. S cannot be placed at a corner
 C. U cannot be placed at a mid-point D. R cannot be placed at a corner

gatecse-2022 quantitative-aptitude geometry triangles two-marks

Answer key

9.57

Venn Diagram (7)

Practice Tests: Test 1 (15Q) Test 2 (3Q)

9.57.1 Venn Diagram: GATE CSE 2010 | Question: 59



25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

- A. 2 B. 17 C. 13 D. 3

gatecse-2010 quantitative-aptitude easy set-theory&algebra venn-diagram 3

Answer key

9.57.2 Venn Diagram: GATE CSE 2016 | Set 2 | GA | Question: 06



Among 150 faculty members in an institute, 55 are connected with each other through Facebook and 85 are connected through Whatsapp. 30 faculty members do not have Facebook or Whatsapp accounts. The numbers of faculty members connected only through Facebook accounts is _____.

- A. 35 B. 45 C. 65 D. 90

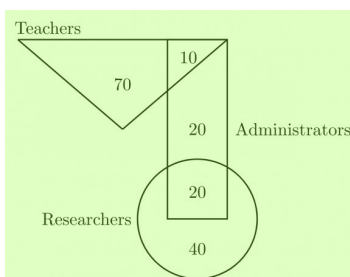
gatecse-2016-set2 quantitative-aptitude venn-diagram easy 7

Answer key

9.57.3 Venn Diagram: GATE CSE 2019 | GA | Question: 7



In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.



10

- A. 0 to 15 B. 16 to 30 C. 31 to 45 D. 46 to 60

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks 12

Answer key

9.57.4 Venn Diagram: GATE CSE 2019 | GA | Question: 9



In a college, there are three student clubs, 60 students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.

- A. 1000 B. 975 C. 900 D. 225

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks 15

Answer key

9.57.5 Venn Diagram: GATE CSE 2024 | Set 2 | GA | Question: 3



In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500.

The number of students who like their core branches is

- A. 1,800 B. 3,500 C. 1,600 D. 1,500

gatecse-2024-set2 quantitative-aptitude venn-diagram one-mark

Answer key

9.57.6 Venn Diagram: GATE2010 TF: GA-8



A gathering of 50 linguists discovered that 4 knew Kannada, Telugu and Tamil, 7 knew only Telugu and Tamil, 5 knew only Kannada and Tamil, 6 knew only Telugu and Kannada. If the number of linguists who knew Tamil is 24 and those who knew Kannada is also 24, how many linguists knew only Telugu?

- A. 9 B. 10 C. 11 D. 8

general-aptitude quantitative-aptitude gate2010-tf venn-diagram 3

Answer key

9.57.7 Venn Diagram: GATE2011 GG: GA-7



In a class of 300 students in an M.Tech programme, each student is required to take at least one subject from the following three:

- M600: Advanced Engineering Mathematics
- C600: Computational Methods for Engineers
- E600: Experimental Techniques for Engineers

The registration data for the M.Tech class shows that 100 students have taken M600, 200 students have taken C600, and 60 students have taken E600. What is the maximum possible number of students in the class who have taken all the above three subjects?

- A. 20 B. 30 C. 40 D. 50

gate2011-gg quantitative-aptitude set-theory&algebra venn-diagram 10

Answer key

9.58

Volume (1)

9.58.1 Volume: GATE CSE 2021 | Set 1 | GA | Question: 6



We have 2 rectangular sheets of paper, M and N, of dimensions $6\text{ cm} \times 1\text{ cm}$ each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____.

- A. $\frac{\pi}{2}$ B. $\frac{3}{\pi}$ C. $\frac{9}{\pi}$ D. 3π

gatecse-2021-set1 quantitative-aptitude mensuration volume two-marks 13

Answer key

9.59

Work Time (3)

Practice Tests: Test 1 (15Q) Test 2 (8Q)

15

9.59.1 Work Time: GATE CSE 2010 | Question: 64



5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long it will take to build the wall?

- A. 20 days B. 18 days C. 16 days D. 15 days

gatecse-2010 quantitative-aptitude normal work-time

Answer key

9.59.2 Work Time: GATE CSE 2011 | Question: 64



A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders.

Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of 5th day? 1

- A. 4 B. 5 C. 6 D. 7 2

gatecse-2011 quantitative-aptitude normal work-time 3

Answer key 

9.59.3 Work Time: GATE CSE 2013 | Question: 65



The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in Rs. is 4

- A. 16,500 B. 15,180 C. 11,000 D. 10,120 5

gatecse-2013 quantitative-aptitude normal work-time 6

Answer key  7

Answer Keys 8

9.1.1	A	9.2.1	D	9.2.2	B	9.2.3	D	9.2.4	B	9
9.2.5	B	9.3.1	B	9.4.1	D	9.4.2	B	9.5.1	A	
9.6.1	C	9.6.2	B	9.6.3	C	9.7.1	C	9.8.1	2006	
9.8.2	C	9.8.3	A	9.8.4	A	9.9.1	B	9.10.1	B	
9.10.2	C	9.10.3	C	9.10.4	C	9.11.1	D	9.11.2	A	
9.11.3	C	9.12.1	A	9.13.1	B	9.13.2	C	9.13.3	D	
9.13.4	C	9.13.5	D	9.14.1	B	9.14.2	D	9.14.3	A	
9.15.1	C	9.15.2	0.4895 : 0.4897	9.16.1	C	9.16.2	C	9.17.1	A	
9.17.2	A	9.17.3	C	9.17.4	1300	9.18.1	D	9.18.2	B	
9.19.1	A	9.20.1	D	9.20.2	A	9.20.3	B	9.21.1	C	
9.22.1	D	9.22.2	C	9.22.3	C	9.23.1	B	9.24.1	8	
9.24.2	C	9.24.3	B	9.24.4	C	9.24.5	B	9.24.6	A	
9.24.7	C	9.25.1	6	9.25.2	A	9.25.3	D	9.25.4	A	
9.25.5	A	9.26.1	A	9.27.1	140	9.27.2	C	9.27.3	C	
9.27.4	B	9.27.5	B	9.28.1	B	9.28.2	C	9.28.3	A	
9.28.4	C	9.28.5	C	9.28.6	D	9.29.1	C	9.30.1	B	
9.30.2	C	9.30.3	C	9.30.4	A	9.31.1	C	9.32.1	D	
9.33.1	B	9.34.1	B	9.34.2	C	9.34.3	C	9.34.4	B	
9.34.5	D	9.35.1	C	9.35.2	A	9.35.3	C	9.35.4	B	
9.36.1	A	9.36.2	X	9.37.1	96	9.37.2	B	9.37.3	D	
9.37.4	B	9.37.5	C	9.37.6	D	9.37.7	A	9.37.8	A	
9.37.9	D	9.38.1	850	9.38.2	D	9.38.3	C	9.38.4	A	
9.38.5	B	9.38.6	D	9.38.7	C	9.38.8	D	9.39.1	A	
9.39.2	B	9.39.3	B	9.39.4	B	9.40.1	32	9.40.2	C	
9.40.3	D	9.40.4	B	9.40.5	D	9.41.1	B	9.42.1	C	
9.43.1	B	9.44.1	D	9.44.2	A	9.44.3	C	9.44.4	D	

9.44.5	C	9.44.6	C	9.44.7	D	9.44.8 ¹	A	9.44.9	C ²
9.44.10	C	9.44.11	A	9.44.12	A	9.44.13	A	9.44.14	C
9.44.15	C	9.44.16	A	9.44.17	25	9.45.1	C	9.45.2	C
9.45.3	A	9.46.1	D	9.46.2	B	9.46.3	B	9.46.4	D
9.46.5	C	9.46.6	B	9.47.1	A	9.48.1	B	9.48.2	B
9.48.3	C	9.48.4	C	9.48.5	D	9.48.6	C	9.48.7	D
9.48.8	B	9.48.9	180	9.49.1	A	9.49.2	B	9.49.3	A
9.49.4	B	9.50.1	B	9.51.1	C	9.51.2	B	9.51.3	C
9.51.4	C	9.52.1	C	9.52.2	A	9.52.3	C	9.53.1	A
9.54.1	A	9.55.1	48	9.55.2	C	9.55.3	C	9.55.4	B
9.55.5	B	9.55.6	C	9.55.7	D	9.56.1	B	9.56.2	A
9.56.3	B	9.57.1	D	9.57.2	A	9.57.3	C	9.57.4	C
9.57.5	A	9.57.6	C	9.57.7	B	9.58.1	C	9.59.1	D
9.59.2	C	9.59.3	B						



Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping, Paper folding, cutting, and patterns in 2 and 3 dimensions

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	0	1	0	0	1	1	2	1	0
2 Marks Count	1	0	1	0	2	1	1	1	0	1	0
Total Marks	3	1	2	1	4	2	3	3	2	3	1

Welcome to the "General Aptitude: Spatial Aptitude" chapter of your GATE Computer Science exam preparation. This section is designed to equip you with the essential concepts, strategies, and quick references needed to excel in spatial reasoning questions. Spatial aptitude tests your ability to visualize, interpret, and manipulate objects and patterns in two and three dimensions. For GATE CS, mastering this area is crucial not just for the General Aptitude section, but also for developing fundamental visualization skills that are indirectly beneficial in areas like computer graphics, data structures, and algorithms. Typically, you can expect 1-3 questions from Spatial Aptitude in the General Aptitude section, contributing 2-5 marks. Questions often involve identifying patterns, mentally rotating shapes, understanding mirror images, or assembling figures, usually presented as Multiple Choice Questions (MCQs).

Topic-wise Key Concepts

1. 3D Structure

Definition and Core Idea: This topic involves understanding and visualizing three-dimensional objects from their two-dimensional representations (like orthographic or isometric views) or vice-versa. It requires the ability to mentally construct, deconstruct, and manipulate 3D shapes, identifying their hidden faces, edges, and vertices.

Important Formulas, Theorems, and Results:

- Euler's Formula for Polyhedra:** For any simple (convex) polyhedron, the number of vertices (V), edges (E), and faces (F) are related by the formula:

$$V - E + F = 2$$

This formula is fundamental for checking the consistency of polyhedral structures.

Common Polyhedra Properties:

- Cube:** $V = 8, E = 12, F = 6$
- Cuboid:** $V = 8, E = 12, F = 6$
- Triangular Prism:** $V = 6, E = 9, F = 5$
- Square Pyramid:** $V = 5, E = 8, F = 5$
- Tetrahedron:** $V = 4, E = 6, F = 4$

Key Properties and Identities:

- Orthographic Projections:** These are 2D views (Front, Top, Side) that show the true shape and size of one face of an object, without perspective distortion.
- Isometric Views:** A 3D representation where all three axes are equally foreshortened, giving a sense of depth and volume. Parallel lines remain parallel.
- Nets of 3D Shapes:** A 2D pattern that can be folded to form a 3D shape. Understanding nets is crucial for visualizing how faces connect.
- Hidden Lines:** In 2D representations of 3D objects, dashed lines often represent edges that are not directly visible from the chosen viewpoint.

Common Pitfalls or Tricky Points:

- Misinterpreting hidden lines or confusing different orthographic views.
- Incorrectly counting faces, edges, or vertices, especially when parts are hidden or joined.
- Difficulty in visualizing the folding/unfolding of complex nets.
- Confusing isometric projection with perspective projection.

Standard Problem-Solving Techniques or Shortcuts:

- **Mental Rotation and Manipulation:** Practice mentally rotating and viewing objects from different angles. 1
- **Feature Tracking:** Focus on a unique corner, edge, or patterned face and track its position across different views or during folding.
- **Elimination Strategy:** Quickly eliminate options that clearly violate geometric principles or don't match specific features.
- **Auxiliary Views:** If struggling, mentally sketch an additional view to clarify the structure.
- **Nets to 3D:** For nets, identify adjacent faces and opposite faces. Opposite faces never share an edge.

2. Assembling 2

Definition and Core Idea: This topic involves combining given 2D or 3D components to form a complete shape or object, or identifying which set of components can form a specified shape. It tests your spatial reasoning to fit parts together precisely without overlap or gaps. 3

Important Formulas, Theorems, and Results: No specific mathematical formulas are typically used. The core idea 4 relies on geometric principles of congruence, tessellation, and conservation of area/volume.

Key Properties and Identities: 5

- **Congruence:** Pieces must match exactly in shape and size to fit together. 6
- **Tessellation:** The ability of shapes to fit together without gaps or overlaps to cover a surface (for 2D assembling).
- **Conservation of Area/Volume:** The total area (for 2D) or volume (for 3D) of the assembled figure must equal the sum of the areas/volumes of its constituent pieces.
- **Edge/Face Matching:** The edges or faces that meet must be identical in length and shape.
- **Symmetry:** Recognizing symmetry in the target shape or individual pieces can aid assembly.

Common Pitfalls or Tricky Points: 7

- Overlooking small details or subtle differences in piece shapes. 8
- Assuming pieces can be resized, distorted, or flipped (unless explicitly allowed).
- Not considering all possible orientations (rotations and reflections) for each piece.
- Difficulty in visualizing how irregular shapes fit into complex outlines.

Standard Problem-Solving Techniques or Shortcuts: 9

- **Start with Distinctive Pieces:** Begin by placing pieces with unique shapes, sharp corners, or specific patterns that 10 have limited fitting options.
- **Work from the Edges/Corners:** For outline-based problems, try to fill the outer boundaries first.
- **Identify Complementary Shapes:** Look for pieces whose contours fit together like puzzle pieces.
- **Mental Rotation and Translation:** Actively rotate and slide pieces mentally to test fits.
- **Elimination:** If a piece clearly doesn't fit in any orientation, eliminate the option containing it.

3. Assembling Pieces 11

Definition and Core Idea: This topic is a specific type of assembling, often focusing on arranging a set of smaller, usually irregular, 2D pieces to form a larger, specified 2D figure. It's akin to solving a jigsaw puzzle or tangram, requiring visualization of how shapes fit together, including rotation and reflection. 12

Important Formulas, Theorems, and Results: No specific formulas. The principles are geometric matching and conservation of area. 13

Key Properties and Identities: 14

- **Area Conservation:** The sum of the areas of the individual pieces must equal the area of the final assembled figure. 15
- **Edge and Vertex Matching:** Edges of adjacent pieces must align perfectly, and vertices must meet precisely.
- **No Overlap or Gaps:** The assembled figure must be solid, with no parts of pieces overlapping and no empty spaces within the boundary.
- **Orientation Flexibility:** Pieces can usually be rotated and/or reflected unless specified otherwise.

Common Pitfalls or Tricky Points: 16

- Incorrectly assuming a piece can be flipped (reflected) when only rotation is allowed, or vice-versa. 17
- Misjudging the size or shape of internal gaps that need to be filled.
- Difficulty in visualizing how multiple irregular pieces combine to form a regular outline.
- Overlooking that pieces might need to be rotated by angles other than 90 degrees.

Standard Problem-Solving Techniques or Shortcuts: 1

- **Focus on Unique Features:** Identify pieces with distinct curves, sharp angles, or long straight edges. 2
- **Largest Piece First:** Often, starting with the largest or most complex piece helps define the initial structure.
- **Work from the Outside In:** Try to form the outer boundary of the target figure first.
- **Consider Negative Space:** Sometimes, it's easier to visualize the shape of the 'hole' that needs to be filled by the remaining pieces.
- **Trial and Error (Mental):** Mentally try fitting pieces, eliminating options as you go.

4. Counting Figure 3

Definition and Core Idea: This involves systematically counting the number of specific geometric shapes (e.g., triangles, squares, rectangles, cubes) within a complex, often overlapping, figure. The core idea is to develop a structured approach to avoid double-counting or missing any figures. 4

Important Formulas, Theorems, and Results: 5

- **Number of Squares in an $n \times n$ Grid:** 6

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad 7$$

For example, in a 3×3 grid, it's $3^2 + 2^2 + 1^2 = 9 + 4 + 1 = 14$ squares. 8

- **Number of Rectangles in an $n \times m$ Grid:** 9

$$\left(\frac{n(n+1)}{2} \right) \times \left(\frac{m(m+1)}{2} \right) \quad 10$$

This counts all possible rectangles, including squares. 11

- **Number of Triangles in a Simple Base with n Segments:** If a base line is divided into n segments and a common vertex is connected to all endpoints, the number of triangles formed is: 12

$$\frac{n(n+1)}{2} \quad 13$$

This applies to triangles sharing a common vertex and lying on the same base line. More complex configurations require specific counting methods. 14

- **Number of Cubes in a Stack:** For a stack of unit cubes, count cubes layer by layer, or by identifying visible and hidden cubes based on the structure. 15

Key Properties and Identities: 16

- **Hierarchical Counting:** Often, figures can be counted by their size (e.g., 1-unit squares, 2x2 squares, etc.) or by the number of smaller units they encompass. 17
- **Systematic Listing:** Labeling vertices or sections can help in systematically listing and counting figures without repetition.
- **Symmetry:** If a figure is symmetrical, counting one part and multiplying can be a shortcut, but be careful with figures on the axis of symmetry.

Common Pitfalls or Tricky Points: 18

- Double-counting figures or missing smaller, embedded figures. 19
- Not having a systematic approach, leading to confusion and errors.
- Difficulty in visualizing overlapping shapes, especially triangles or rectangles.
- For 3D figures, forgetting to count hidden cubes or structures.

Standard Problem-Solving Techniques or Shortcuts: 20

- **Size-Based Counting:** Count all figures of the smallest size, then combinations of two, three, and so on, up to the largest possible figure. 21
- **Vertex/Line-Based Counting:** For triangles, identify a common vertex and count the number of base segments. For rectangles, count horizontal and vertical lines.
- **Decomposition:** Break down complex figures into simpler, countable components.

- **Labeling:** Assign numbers or letters to distinct regions or vertices to help track counted figures. 1
- **Pattern Recognition:** For grids, apply the formulas directly.

5. Grouping 2

Definition and Core Idea: This topic involves identifying a common characteristic or underlying rule among a set of figures and grouping them accordingly, or identifying the 'odd one out' that does not share the common property. It primarily tests pattern recognition and classification based on visual attributes. 3

Important Formulas, Theorems, and Results: No specific formulas. Relies purely on observational skills and logical deduction. 4

Key Properties and Identities: 5

- **Number of Elements/Sides:** Counting internal components, sides of a polygon, or lines. 6
- **Symmetry:** Presence or absence of rotational, reflectional, or point symmetry.
- **Orientation:** The direction or alignment of figures or their internal components.
- **Shading/Filling Patterns:** Whether parts are filled, hollow, or have specific textures.
- **Relative Positions:** The spatial relationship between different elements within a figure (e.g., inside/outside, left/right).
- **Transformations:** Whether figures are related by rotation, reflection, or translation.
- **Connectivity:** Whether elements are connected or separate.

Common Pitfalls or Tricky Points: 7

- Focusing on superficial differences rather than fundamental commonalities. 8
- Missing subtle patterns or rules that apply to most figures.
- Not considering all possible attributes (e.g., only looking at shape, not shading or orientation).
- Assuming a simple rule when a more complex one is at play.

Standard Problem-Solving Techniques or Shortcuts: 9

- **Attribute Listing:** For each figure, list all observable attributes (e.g., number of sides, type of lines, shading, orientation of internal arrow). 10
- **Compare and Contrast:** Systematically compare each figure with the others to find commonalities or unique features.
- **Hypothesis Testing:** Formulate a hypothesis about a potential grouping rule and test it against all figures.
- **Look for Deviations:** When finding the odd one out, identify the property that one figure lacks or possesses uniquely.
- **Consider Transformations:** Check if figures can be grouped based on being rotations or reflections of each other.

6. Image Rotation 11

Definition and Core Idea: This topic involves mentally rotating a 2D or 3D object to match a given orientation or to predict its appearance after a specified rotation. It tests your ability to visualize how shapes and their internal components change position relative to each other during rotation. 12

Important Formulas, Theorems, and Results: No specific formulas, but understanding angular rotation (e.g., 90°, 180°, 270° clockwise/anticlockwise) is key. 13

Key Properties and Identities: 14

- **Relative Positions:** The spatial relationship between different parts of the object remains constant during rotation. Only the object's overall orientation changes. 15
- **Clockwise vs. Anticlockwise:** Distinguish between the direction of rotation.
- **Degrees of Rotation:** Standard rotations are often multiples of 90 degrees. A 180-degree rotation results in an inverted image (upside down). A 360-degree rotation returns the object to its original orientation.
- **Fixed Point/Axis:** Rotation occurs around a central point (for 2D) or an axis (for 3D).

Common Pitfalls or Tricky Points: 16

- Incorrectly rotating parts of the image independently instead of the whole figure. 17
- Confusing clockwise with anticlockwise rotation.
- Misjudging the exact angle of rotation, especially for non-90-degree rotations.
- Difficulty in tracking multiple features simultaneously during rotation.

Standard Problem-Solving Techniques or Shortcuts: 1

- **Focus on a Distinctive Feature:** Pick a unique part of the figure (e.g., an arrow, a specific corner, a shaded region) and track its movement during rotation. 2
- **Mental Overlay:** Imagine overlaying the rotated figure onto the original to check alignment.
- **Use Reference Points:** Mentally fix a point on the figure and observe its path during rotation.
- **Elimination:** Quickly rule out options that are clearly not rotations (e.g., reflections, altered shapes).
- **Practice with Physical Objects:** If struggling, use a physical object (like a matchbox or a drawn shape on paper) and rotate it to build intuition.

7. Mirror Image 3

Definition and Core Idea: This topic involves determining the appearance of an object or figure when reflected across a mirror line (which can be vertical, horizontal, or diagonal). The core idea is to understand the concept of lateral inversion. 4

Important Formulas, Theorems, and Results: No specific formulas. The principle is geometric reflection. 5

Key Properties and Identities: 6

- **Lateral Inversion (Vertical Mirror):** For a vertical mirror, left becomes right, and right becomes left. The top and bottom parts remain in their respective positions. 7
- **Vertical Inversion (Horizontal Mirror):** For a horizontal mirror, top becomes bottom, and bottom becomes top. The left and right parts remain in their respective positions.
- **Distance Preservation:** Every point in the original figure is the same perpendicular distance from the mirror line as its corresponding point in the mirror image.
- **Symmetry:** If a figure has reflectional symmetry along the mirror line, its mirror image will be identical to the original.
- **Text/Numbers:** Letters and numbers appear reversed in a mirror image (e.g., 'b' becomes 'd', 'p' becomes 'q', '6' becomes '9'). Some letters (A, H, I, M, O, T, U, V, W, X, Y) are symmetrical and appear unchanged in a vertical mirror.

Common Pitfalls or Tricky Points: 8

- Confusing mirror image with rotation (especially 180-degree rotation, which looks similar to a horizontal mirror image followed by a vertical mirror image). 9
- Incorrectly reflecting across diagonal mirror lines, which requires more complex visualization.
- Misinterpreting internal patterns or text within the figure.
- Forgetting that the 'depth' or 'front-back' orientation is also inverted in a 3D mirror image.

Standard Problem-Solving Techniques or Shortcuts: 10

- **Imagine Folding:** Mentally fold the paper along the mirror line. The image should appear on the other side. 11
- **Point-by-Point Reflection:** For complex figures, reflect key points or vertices one by one.
- **Feature Tracking:** Pick a distinctive feature (e.g., an arrow, a shaded corner) and visualize its reflected position.
- **Elimination:** Rule out options that show rotation or distortion instead of reflection.
- **Practice with Letters/Numbers:** Familiarize yourself with the mirror images of common letters and digits.

8. Paper Folding 12

Definition and Core Idea: This topic involves visualizing a sequence of folds made on a piece of paper and then predicting the pattern of holes or cuts when the paper is unfolded, or vice-versa. The core idea is to mentally reverse the folding process, understanding how cuts/holes multiply and reflect across each fold line. 13

Important Formulas, Theorems, and Results: No specific formulas. Relies on principles of symmetry and reflection. 14

Key Properties and Identities:

- **Symmetry:** Each fold line acts as a mirror line. When the paper is unfolded, any cut or hole will be reflected across that fold line. 15
- **Number of Layers:** A cut made through multiple layers of folded paper will result in multiple holes upon unfolding, corresponding to the number of layers cut.
- **Order of Unfolding:** The paper must be unfolded in the reverse order of folding. The last fold made is the first one to be unfolded.
- **Position of Cuts:** The position of a cut relative to the edges and previous fold lines is crucial.

Common Pitfalls or Tricky Points: 16

- Incorrectly reversing the order of folds during unfolding.
- Missing reflections across fold lines, especially when multiple folds are involved.
- Misjudging the number of layers a cut passes through.
- Difficulty in visualizing the exact shape and position of holes after multiple reflections.
- Confusing the orientation of the paper after each fold.

1

Standard Problem-Solving Techniques or Shortcuts: 2

- **Work Backward:** Start from the final folded and cut state and mentally unfold the paper step-by-step, applying reflection at each fold line.
- **Draw Each Stage:** If the problem is complex, quickly sketch the paper at each stage of unfolding, marking the reflections of the cuts.
- **Identify Fold Lines as Mirror Lines:** Each fold line becomes an axis of symmetry for the cuts/holes when unfolded.
- **Focus on a Single Cut/Hole:** Track how one specific cut or hole multiplies and reflects across the folds.
- **Eliminate Options:** Use the first few unfolding steps to eliminate options that don't match the expected pattern.

3

9. Patterns In Three Dimensions 4

Definition and Core Idea: This topic involves identifying sequences or transformations of 3D objects, predicting the next object in a series, or identifying relationships between 3D patterns. It extends 2D pattern recognition skills to three dimensions, often involving rotation, translation, and changes in internal features of 3D shapes.

5

Important Formulas, Theorems, and Results: No specific formulas. Relies on understanding 3D geometry and transformations.

6

Key Properties and Identities: 7

- **3D Transformations:**
 - **Rotation:** Around X, Y, or Z axes (e.g., a cube rotating on its edge or face).
 - **Translation:** Movement along an axis.
 - **Scaling:** Change in size (less common in GATE spatial aptitude).
- **Internal Features:** Patterns, markings, or smaller objects within the 3D structure can also follow a sequence of changes.
- **Structural Changes:** Addition or removal of smaller cubes or components from a larger 3D structure.
- **Perspective Changes:** How the 2D representation of a 3D object changes with different viewpoints.

9

Common Pitfalls or Tricky Points: 10

- Difficulty in visualizing complex 3D changes from static 2D representations.
- Missing subtle transformations or changes in internal elements.
- Confusing rotations around different axes.
- Overlooking hidden changes in the back or underside of the 3D object.

11

Standard Problem-Solving Techniques or Shortcuts: 12

- **Focus on One Aspect at a Time:** Analyze the pattern of change for one specific feature (e.g., the top face, a specific corner, an internal dot) before combining observations.
- **Mentally Manipulate:** Actively imagine the 3D objects moving or transforming.
- **Look for Consistent Rules:** Identify a clear, repeatable rule or sequence of operations that applies to all given figures in the series.
- **Use a Reference Point/Axis:** Mentally fix a point or an axis on the object to track its movement or rotation.
- **Elimination:** Rule out options that clearly do not follow the established pattern.

13

10. Patterns In Two Dimensions 14

Definition and Core Idea: This topic involves identifying the underlying rule or sequence in a series of 2D figures and predicting the next figure, or finding the odd one out. It requires recognizing transformations (rotation, reflection, translation), changes in number, size, shading, or arrangement of elements within the figures.

15

Important Formulas, Theorems, and Results: No specific formulas. Relies on logical deduction and visual pattern recognition.

16

Key Properties and Identities: 17

- **Transformations:**
 - **Rotation:** Clockwise or anticlockwise, by specific degrees (e.g., 45°, 90°, 180°).

18

- **Reflection:** Across a horizontal, vertical, or diagonal axis.
- **Translation:** Movement of the entire figure or its components in a specific direction.

1

• **Changes in Attributes:** 2

- **Number:** Increase or decrease in the count of elements.
- **Size/Shape:** Enlargement, reduction, or alteration of shapes.
- **Shading/Filling:** Alternating or progressive changes in shading.
- **Arrangement:** Changes in the relative positions of components.
- **Connectivity:** Elements joining or separating.

3

• **Alternating Patterns:** A pattern that alternates between two or more states. 4

• **Progressive Patterns:** A pattern that changes incrementally with each step.

Common Pitfalls or Tricky Points: 6

- Overlooking subtle changes in size, orientation, or shading.
- Assuming a simple pattern when a more complex, multi-layered rule is present.
- Not considering all possible transformations (e.g., only looking for rotation, ignoring reflection).
- Getting confused by multiple elements changing simultaneously.

7

Standard Problem-Solving Techniques or Shortcuts: 8

- **Systematic Analysis:** Break down the figure into its constituent elements. Analyze how each element (e.g., outer shape, inner shape, arrow, dot, shading) changes from one figure to the next.
- **Track One Element at a Time:** If multiple elements are changing, focus on one element's transformation throughout the series before moving to the next.
- **Look for Cycles:** Identify if a pattern repeats after a certain number of figures.
- **Identify the Rule:** Formulate a clear rule (e.g., "the inner square rotates 90 degrees clockwise, and the outer circle alternates shading").
- **Test the Rule:** Apply your hypothesized rule to the subsequent figures in the series to confirm its validity.
- **Elimination:** Use the identified pattern to quickly eliminate options that do not conform.

9

Quick Formula Reference 10

This section consolidates the key formulas and mathematical results important for Spatial Aptitude problems. 11

- **Euler's Formula for Polyhedra:** For any simple polyhedron, the relationship between vertices (V), edges (E), and faces (F) is given by: 12

$$V - E + F = 2 \quad 13$$

- **Number of Squares in an $n \times n$ Grid:** The total number of squares (of all sizes) in an $n \times n$ grid is: 14

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad 15$$

- **Number of Rectangles in an $n \times m$ Grid:** The total number of rectangles (including squares) in an $n \times m$ grid is: 16

$$\left(\frac{n(n+1)}{2} \right) \times \left(\frac{m(m+1)}{2} \right) \quad 17$$

- **Number of Triangles in a Simple Base with n Segments:** If a base line is divided into n segments and a common vertex is connected to all endpoints, the number of triangles formed is: 18

$$\frac{n(n+1)}{2} \quad 19$$

Important Tips for GATE 20

Mastering Spatial Aptitude for GATE requires not just understanding concepts but also developing effective exam strategies. Here are some practical tips: 21

1. **Visualize Actively, Don't Just Observe:** Spatial aptitude is about mental manipulation. Don't just look at the figures; actively imagine them moving, rotating, folding, or reflecting. Close your eyes briefly if it helps to form a clearer mental image. 22

2. **Adopt a Systematic Approach:** Avoid haphazard guessing. For each problem type (e.g., counting, paper folding, pattern series), develop and follow a structured method. This prevents errors like double-counting or missing steps.
3. **Focus on Key Features and Reference Points:** When dealing with rotations, reflections, or complex patterns, identify a unique or distinctive element (e.g., a specific corner, an arrow, a shaded region). Track its movement or transformation throughout the sequence or across options.
4. **Leverage the Elimination Strategy:** In multiple-choice questions, often several options can be quickly ruled out because they clearly violate a fundamental rule (e.g., a reflection instead of a rotation, an extra piece in an assembly). This narrows down your choices and increases your chances.
5. **Practice Under Time Constraints:** Spatial aptitude questions can be time-consuming due to the mental effort required. Practice solving problems within strict time limits to improve your speed and efficiency. The more you practice, the faster your visualization skills will become.
6. **Clearly Differentiate Mirror Image from Rotation:** These are common sources of confusion. Remember that a mirror image involves lateral inversion (left-right flip for a vertical mirror), while rotation changes orientation without flipping. A 180-degree rotation can sometimes look similar to a mirror image, so pay close attention.
7. **Don't Hesitate to Sketch (Mentally or Physically):** For complex problems like paper folding or 3D structures, a quick mental or physical sketch of intermediate steps can significantly clarify the problem and prevent errors. This is especially useful for tracking reflections in paper folding.
8. **Start with Simpler Problems to Build Confidence:** If you find a particular type of spatial problem challenging, begin with simpler variations to build your foundational understanding and confidence before tackling more complex ones. Gradually increase the difficulty.

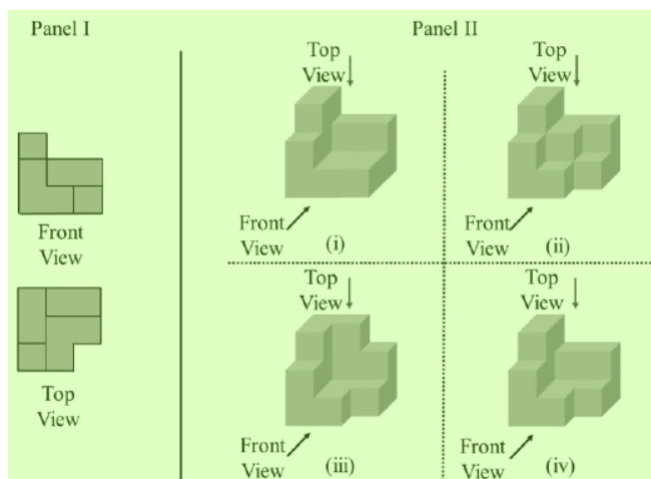
10.1

3d Structure (1)

10.1.1 3d Structure: GATE CSE 2026 | Set 1 | GA | Question: 7



In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I?



- A. (i) B. (ii) C. (iii) D. (iv)

gates-2026-set1 spatial-aptitude 3d-structure two-marks 7

Answer key

10.2

Assembling (1)

10.2.1 Assembling: GATE DA 2026 | GA | Question: 3



The four pieces of a puzzle are shown in the figure below.

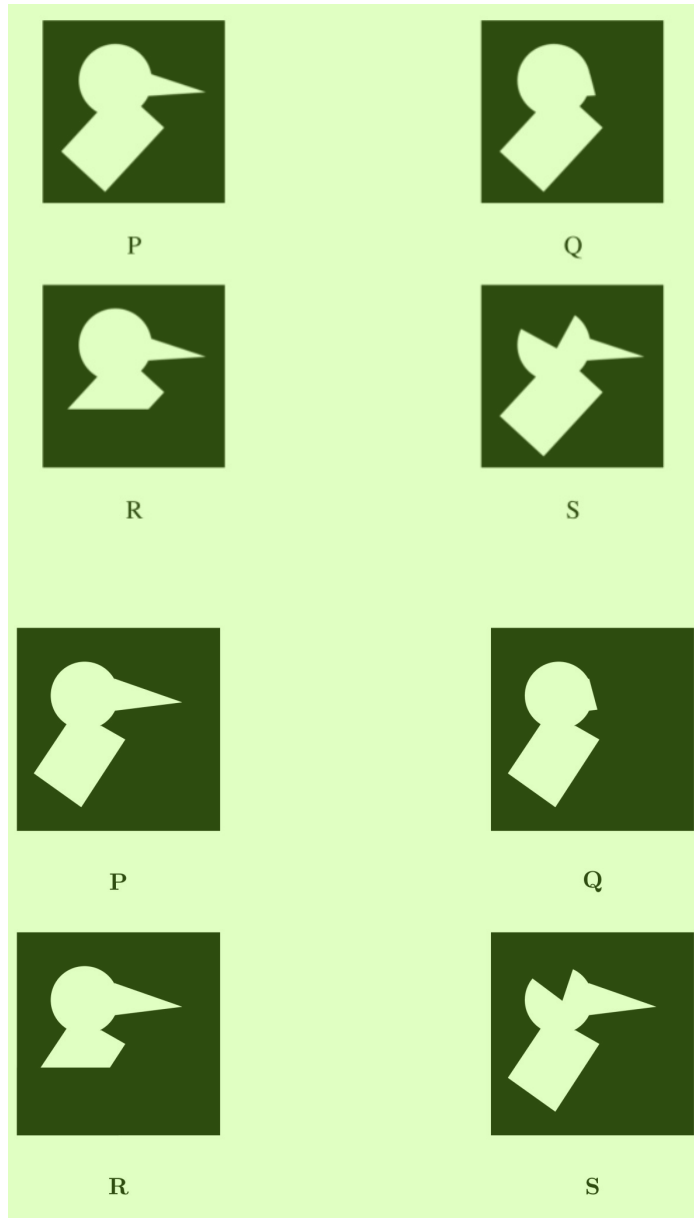




1

Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?

2



3

A. P B. Q C. R D. S

4

gateda-2026 spatial-aptitude assembling one-mark

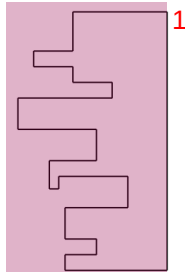
Answer key

10.3

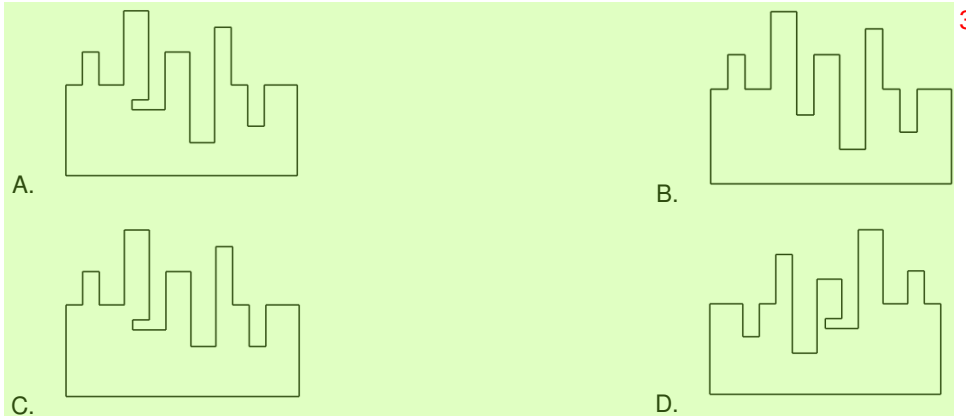
Assembling Pieces (1)

10.3.1 Assembling Pieces: GATE CSE 2021 | Set 2 | GA | Question: 7





A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.



gatecse-2021-set2 spatial-aptitude assembling-pieces two-marks 4

Answer key 5

10.4 Image Rotation (3)

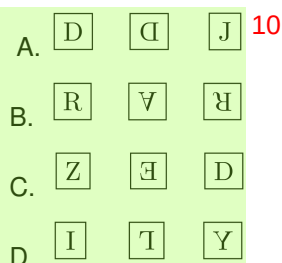
Practice Test: Test 1 (7Q) 6

10.4.1 Image Rotation: GATE CSE 2022 | GA | Question: 5

A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters.



From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.



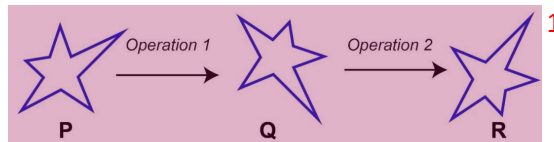
gatecse-2022 spatial-aptitude image-rotation one-mark

Answer key

10.4.2 Image Rotation: GATE CSE 2023 | GA | Question: 10

Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?





- A. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- B. Operation 1: A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- C. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line
- D. Operation 1: A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line

gatecse-2023 spatial-aptitude image-rotation two-marks

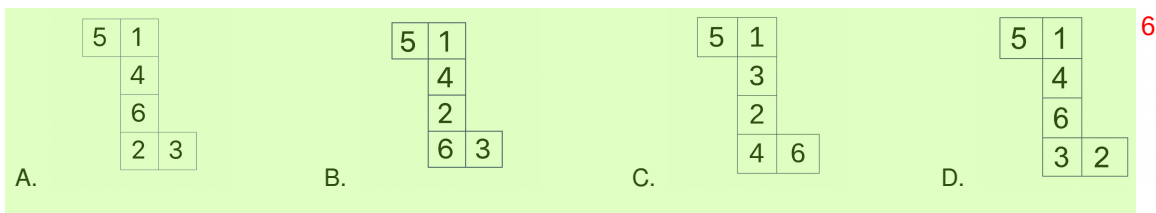
Answer key

10.4.3 Image Rotation: GATE DS&AI 2024 | GA | Question: 9

Three different views of a dice are shown in the figure below.



The piece of paper that can be folded to make this dice is



gate-ds-ai-2024 spatial-aptitude image-rotation two-marks

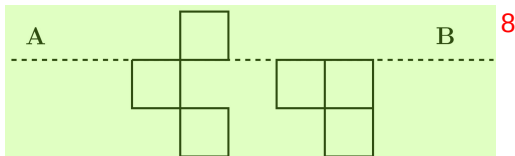
Answer key

10.5 Mirror Image (1)

Practice Test: Test 1 (10Q)

10.5.1 Mirror Image: GATE CSE 2024 | Set 1 | GA | Question: 10

The least number of squares to be added in the figure to make A B a line of symmetry is



- A. 6 B. 4 C. 5 D. 7

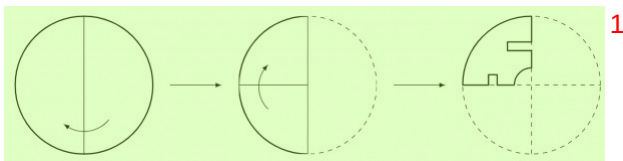
gatecse-2024-set1 spatial-aptitude mirror-image two-marks

Answer key

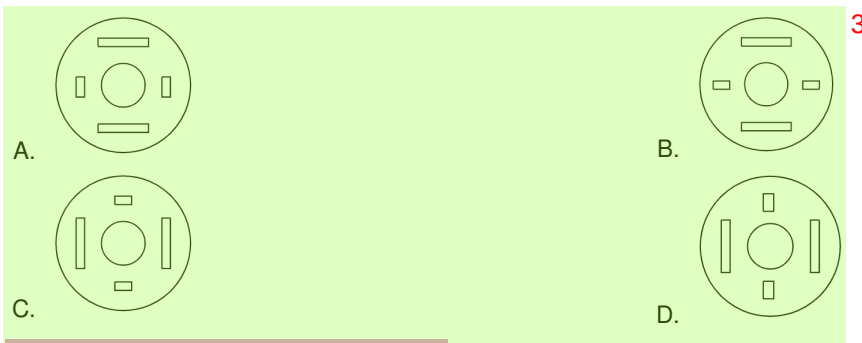
10.6 Paper Folding (5)

Practice Tests: Test 1 (15Q) Test 2 (5Q)

10.6.1 Paper Folding: GATE CSE 2021 | Set 1 | GA | Question: 4



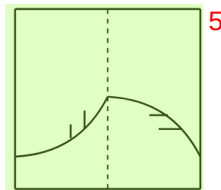
A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.



gatecse-2021-set1 spatial-aptitude paper-folding one-mark

Answer key

10.6.2 Paper Folding: GATE CSE 2021 | Set 2 | GA | Question: 2



A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.



gatecse-2021-set2 spatial-aptitude paper-folding one-mark

Answer key

10.6.3 Paper Folding: GATE CSE 2024 | Set 1 | GA | Question: 9



A rectangular paper of $20 \text{ cm} \times 8 \text{ cm}$ is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

- A. 18 B. 24 C. 20 D. 21

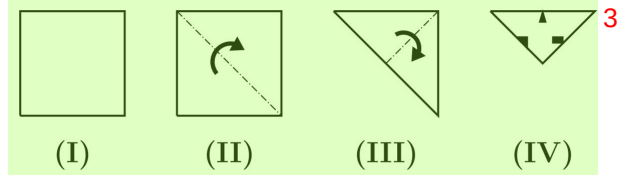
gatecse-2024-set1 spatial-aptitude paper-folding two-marks

10.6.4 Paper Folding: GATE CSE 2025 | Set 1 | GA | Question: 9



A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

Note: The figures shown are representative. 2



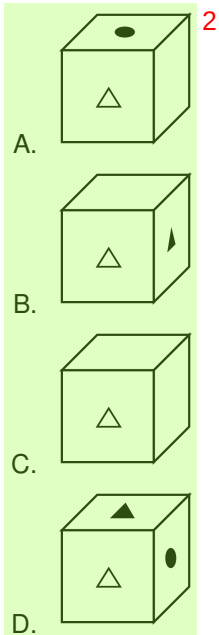
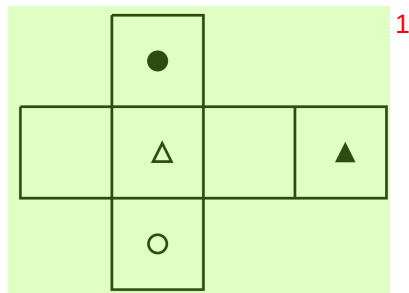
- A. 4
- B.
- C.
- D.

10.6.5 Paper Folding: GATE CSE 2025 | Set 2 | GA | Question: 4



The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.



gatecse2025-set2 spatial-aptitude paper-folding easy one-mark

Answer key

10.7

Patterns In Three Dimensions (3)

Practice Test: Test 1 (10Q) 3

10.7.1 Patterns In Three Dimensions: GATE CSE 2023 | GA | Question: 5

Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE? 4

- A. The surface of the object must be concave everywhere.
- B. The surface of the object must be convex everywhere.
- C. The surface of the object may be concave in some places and convex in other places.
- D. The object can have edges, but no corners.

gatecse-2023 spatial-aptitude patterns-in-three-dimensions one-mark

Answer key

10.7.2 Patterns In Three Dimensions: GATE CSE 2024 | Set 2 | GA | Question: 9

A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

- A. 3
- B. 4
- C. 7
- D. 8

Answer key

10.7.3 Patterns In Three Dimensions: GATE DS&AI 2024 | GA | Question: 10

Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

- A. A rhombus B. A triangle C. An ellipse D. A hexagon ²

Answer key ⁴

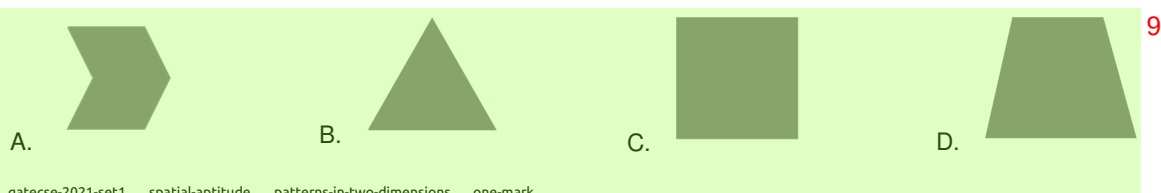
10.8 Patterns In Two Dimensions (4)

Practice Test: Test 1 (10Q) ⁵

10.8.1 Patterns In Two Dimensions: GATE CSE 2021 | Set 1 | GA | Question: 2

A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.

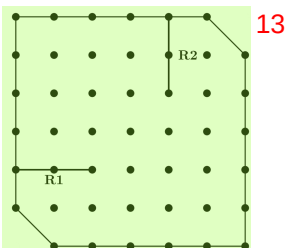
Which one of the following is NOT a convex polygon? ⁸

Answer key ¹⁰

10.8.2 Patterns In Two Dimensions: GATE CSE 2022 | GA | Question: 10

A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved.

What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line. ¹²



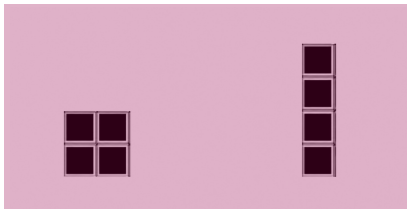
- A. 2 B. 4 C. 5 D. 3 ¹⁴

Answer key

10.8.3 Patterns In Two Dimensions: GATE CSE 2026 | Set 1 | GA | Question: 2

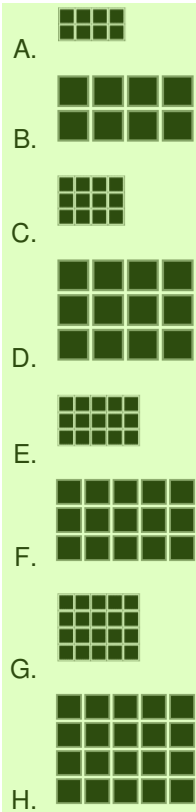
The figure shows two 4-tile patterns.





1

Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below cannot be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?



3

gatecse-2026-set1 spatial-aptitude patterns-in-two-dimensions one-mark

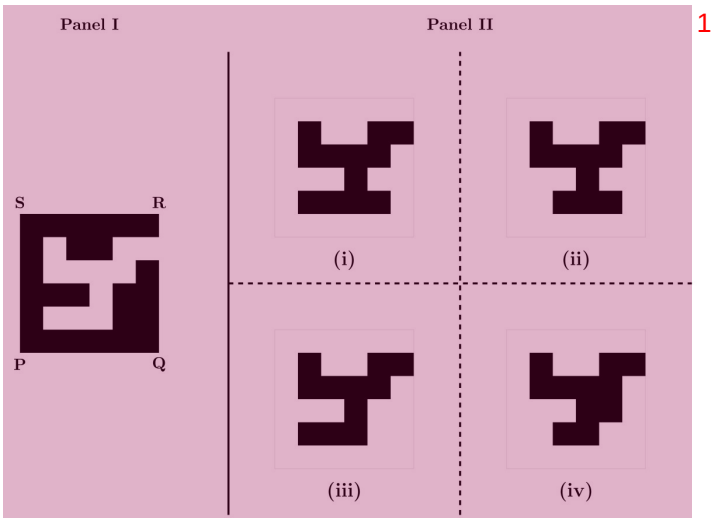
Answer key

10.8.4 Patterns In Two Dimensions: GATE CSE 2026 | Set 2 | GA | Question: 2

A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part?



4



A. (i) B. (ii) C. (iii) D. (iv) 2

gatecse-2026-set2 spatial-aptitude one-mark patterns-in-two-dimensions

Answer key

Answer Keys 3

10.1.1	D
10.4.3	A
10.6.4	A
10.8.1	A

10.2.1	A
10.5.1	A
10.6.5	A
10.8.2	D

10.3.1	A
10.6.1	A
10.7.1	C
10.8.3	C

10.4.1	B
10.6.2	B
10.7.2	A
10.8.4	C

10.4.2	A
10.6.3	A
10.7.3	A



Syllabus: English grammar, Sentence completion. Verbal analogies, Word groups. Instructions, Critical reasoning and Verbal deduction **1**

Mark Distribution in Previous GATE **2**

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum 3
1 Mark Count	1	1	2	2	1	1	2	1	2	2	1
2 Marks Count	0	0	1	0	1	1	1	1	1	1	0
Total Marks	1	1	4	2	3	3	4	3	4	4	1

Welcome to the "General Aptitude: Verbal Aptitude" chapter of your GATE Computer Science exam preparation. This section is designed to equip you with the essential linguistic and reasoning skills necessary to excel in the verbal aptitude segment of the GATE exam. Verbal Aptitude assesses your comprehension of English language, grammar, vocabulary, and logical reasoning abilities, which are crucial not just for the exam but also for effective communication in your professional career. Typically, verbal aptitude questions constitute a significant portion (around 5-8 marks out of 15 for General Aptitude) and can range from direct vocabulary and grammar questions to more complex reading comprehension and logical deduction problems. Mastering these topics can significantly boost your overall GATE score. **4**

Topic-wise Key Concepts **5**

Antonyms **6**

Definition and core idea: Antonyms are words that have opposite meanings. Understanding antonyms enhances vocabulary and comprehension by highlighting contrasts in meaning. **7**

• Key properties and identities: **8**

- Antonyms can be absolute (e.g., hot/cold) or relative (e.g., happy/sad). **9**
- Often formed by adding prefixes (e.g., *un-*, *dis-*, *in-*, *im-*, *non-*) to a word.
- Context is crucial; a word can have different antonyms depending on its usage.

• Common pitfalls: **10**

- Choosing a word that is merely different, not opposite. **11**
- Ignoring prefixes/suffixes that might indicate an antonym.
- Misinterpreting the nuance of the given word.

• Standard problem-solving techniques: **12**

- Understand the exact meaning of the given word. **13**
- Look for prefixes that reverse meaning (e.g., *capable* -> *incapable*).
- Consider the context if the word has multiple meanings.
- Eliminate options that are synonyms or unrelated.
- If unsure, try to use the options in a sentence to see which one creates an opposite meaning.

Articles **14**

Definition and core idea: Articles are words (a, an, the) that define whether a noun is specific or unspecific. They are determiners that precede nouns. **15**

• Important formulas, theorems, and results: **16**

- Indefinite Article 'a':** Used before singular, countable nouns that begin with a consonant sound. E.g., *a book*, *a university* (starts with 'y' sound). **17**
- Indefinite Article 'an':** Used before singular, countable nouns that begin with a vowel sound. E.g., *an apple*, *an hour* (starts with 'o' sound).
- Definite Article 'the':** Used before specific or unique nouns, or nouns that have been previously mentioned. E.g., *the sun*, *the book I read yesterday*.
- Zero Article:** No article is used before plural countable nouns, uncountable nouns, proper nouns, or abstract nouns when speaking generally. E.g., *water is essential*, *students learn*.

• Key properties and identities: **18**

- 'A' and 'an' are used for general or non-specific items. **19**
- 'The' is used for specific or previously identified items.
- Sound, not spelling, determines 'a' vs. 'an'.

• Common pitfalls: **20**

- Confusing 'a' and 'an' based on spelling rather than sound.

- Overusing 'the' with general nouns or proper nouns. 1
- Omitting articles where they are required for singular countable nouns.

• Standard problem-solving techniques: 2

1. Identify if the noun is singular/plural, countable/uncountable. 3
2. Determine if the noun is specific or general in the context.
3. Pay attention to the initial sound of the word following the article.
4. Check for common phrases or idioms that require specific article usage.

Comparative Forms 4

Definition and core idea: Comparative forms of adjectives and adverbs are used to compare two things, while 5
superlative forms are used to compare three or more things.

• Important formulas, theorems, and results: 6

1. **One-syllable adjectives/adverbs:** Add *-er* for comparative, *-est* for superlative. E.g., *tall* → *taller* → *tallest*. 7
2. **Two-syllable adjectives ending in -y:** Change *-y* to *-i* and add *-er/-est*. E.g., *happy* → *happier* → *happiest*.
3. **Two or more syllable adjectives/adverbs:** Use *more* for comparative, *most* for superlative. E.g., *beautiful* → *more beautiful* → *most beautiful*.
4. **Irregular forms:** Some words have irregular comparative/superlative forms. E.g., *good* → *better* → *best*; *bad* → *worse* → *worst*; *much/many* → *more* → *most*.
5. **Comparison structure:** Comparative forms are often followed by *than*. E.g., *X is taller than Y*. 8

• Key properties and identities: 9

- Do not use double comparatives/superlatives (e.g., "more happier" is incorrect).
- Ensure logical comparison (comparing like with like).

• Common pitfalls: 10

- Using "more" or "most" with *-er* or *-est* endings. 11
- Incorrectly forming irregular comparatives/superlatives.
- Making illogical comparisons (e.g., "My car is faster than John").

• Standard problem-solving techniques: 12

1. Count the syllables of the adjective/adverb. 13
2. Check for irregular forms.
3. Ensure the comparison is grammatically and logically sound.

English Grammar 14

Definition and core idea: English grammar refers to the set of structural rules governing the composition of clauses, 15
phrases, and words in the English language. It ensures clarity and correctness in communication.

• Important formulas, theorems, and results: 16

1. **Basic Sentence Structure:** *Subject + Verb + Object* (SVO) is the most common structure. E.g., *She reads a book*. 17
2. **Parts of Speech:** Nouns, Pronouns, Verbs, Adjectives, Adverbs, Prepositions, Conjunctions, Interjections. Each has specific roles.
3. **Subject-Verb Agreement:** The verb must agree in number with its subject. E.g., *He runs* (singular); *They run* (plural).
4. **Tense Consistency:** Maintain a consistent tense throughout a sentence or paragraph unless there's a specific reason to change.
5. **Parallelism:** Similar items in a series or comparison should be in the same grammatical form. E.g., *Helikes swimming, hiking, and cycling*.

• Key properties and identities: 18

- Grammar provides the framework for meaningful sentences.
- Correct grammar prevents ambiguity and misinterpretation.

• Common pitfalls: 19

- Errors in subject-verb agreement, pronoun agreement, and tense usage. 20
- Lack of parallelism.
- Misplaced modifiers.
- Incorrect punctuation.

• Standard problem-solving techniques: 21

1. Break down the sentence into its core components (subject, verb, object). 22
2. Identify the part of speech for each word.
3. Check for agreement (subject-verb, pronoun-antecedent).

4. Look for consistency in tense and structure. 1
5. Read the sentence aloud to catch awkward phrasing or obvious errors.

Grammatical Error 2

Definition and core idea: Grammatical errors are mistakes in the structure or usage of language that violate standard English grammar rules. Identifying and correcting them is key to effective communication. 3

• Key properties and identities: 4

- Common error types include subject-verb agreement, pronoun agreement, tense consistency, parallelism, misplaced modifiers, comma splices, run-on sentences, fragment sentences, and incorrect word choice. 5

• Common pitfalls: 6

- Overlooking subtle errors, especially in complex sentences.
- Focusing only on obvious errors and missing nuanced ones.
- Not knowing the specific rule being tested.

• Standard problem-solving techniques: 7

1. Read the sentence carefully, looking for any awkward phrasing. 8
2. Check for subject-verb agreement (singular subject, singular verb; plural subject, plural verb).
3. Verify pronoun-antecedent agreement (pronoun matches the noun it refers to in number and gender).
4. Ensure tense consistency throughout the sentence.
5. Look for parallel structure in lists or comparisons.
6. Identify misplaced or dangling modifiers.
7. Check for correct use of articles, prepositions, and conjunctions.
8. Consider punctuation errors.

Incorrect Sentence Part 9

Definition and core idea: This question type presents a sentence divided into several parts, one of which contains a grammatical error. The task is to identify the erroneous part. 10

• Key properties and identities: 11

- Errors can span any grammatical rule: subject-verb agreement, tense, pronouns, articles, prepositions, parallelism, vocabulary, etc. 12
- Often, the error is subtle and requires careful attention to detail.

• Common pitfalls: 13

- Rushing through the sentence and missing the error.
- Fixating on one type of error and ignoring others.
- Assuming a part is correct because it "sounds" okay.

• Standard problem-solving techniques: 14

1. Read the entire sentence first to understand its meaning and flow. 15
2. Examine each part individually, checking for common grammatical errors: 16
 - **Subject-Verb Agreement:** Does the verb match the subject in number?
 - **Tense:** Is the tense consistent and appropriate for the context?
 - **Pronouns:** Do pronouns agree with their antecedents in number, gender, and case?
 - **Articles/Prepositions:** Are they used correctly?
 - **Parallelism:** Are items in a list or comparison structured similarly?
 - **Word Choice/Vocabulary:** Is the correct word used (e.g., affect vs. effect)?
 - **Modifiers:** Are adjectives and adverbs placed correctly?
3. If no error is found, consider the "No Error" option if available. 17

Most Appropriate Word 18

Definition and core idea: This involves choosing the word that best fits the meaning and context of a given sentence or passage from a set of options. 19

• Key properties and identities: 20

- Requires strong vocabulary and understanding of nuances in word meanings. 21
- Context is paramount; a word might be grammatically correct but semantically inappropriate.

• Common pitfalls: 22

- Choosing a word that is a near-synonym but doesn't fit the exact context. 23
- Ignoring the tone or register of the sentence.
- Misinterpreting the overall meaning of the sentence.

• Standard problem-solving techniques: 1

1. Read the sentence carefully to grasp its full meaning and tone. 2
2. Consider the part of speech required for the blank (noun, verb, adjective, adverb).
3. Substitute each option into the sentence.
4. Evaluate which option makes the most sense semantically and grammatically.
5. Look for collocations (words that naturally go together) or idiomatic expressions.
6. Eliminate options that are clearly incorrect in meaning or grammar.

Narrative Sequencing 3

Definition and core idea: This task involves arranging a set of jumbled sentences or paragraphs into a coherent and logical narrative sequence. 4

• Key properties and identities: 5

- A coherent narrative follows a logical progression, often chronological, cause-and-effect, or problem-solution. 6
- Transition words and pronouns are crucial clues.

• Common pitfalls: 7

- Ignoring transition words or pronoun references.
- Focusing only on the first sentence and not the overall flow.
- Assuming a chronological order when a different logical order is intended.

• Standard problem-solving techniques: 8

1. **Identify the opening sentence:** Look for sentences that introduce a topic, character, or setting, often without pronouns or conjunctions referring to previous sentences. 9
2. **Identify transition words/phrases:** Words like *however, therefore, meanwhile, subsequently, in addition, finally*, etc., indicate relationships between sentences. 10
3. **Look for pronoun references:** A pronoun (he, she, it, they, this, that) must refer back to a noun introduced in a previous sentence. 11
4. **Establish cause-and-effect or chronological order:** Determine if events follow a time sequence or if one event leads to another. 12
5. **Identify the concluding sentence:** Look for sentences that summarize or provide a final thought. 13
6. **Form pairs:** Try to identify two sentences that logically follow each other. 14
7. **Read the assembled sequence:** Once you have an order, read it through to ensure it flows logically and makes sense. 15

Noun Verb Adjective 16

Definition and core idea: This topic focuses on understanding the definitions, functions, and interrelationships of nouns (naming words), verbs (action/state words), and adjectives (describing words). 17

• Important formulas, theorems, and results: 18

1. **Noun:** Names a person, place, thing, or idea. Can be subject or object. E.g., *student, city, book, freedom*. 19
2. **Verb:** Expresses an action, occurrence, or state of being. E.g., *run, become, is*.
3. **Adjective:** Modifies or describes a noun or pronoun. Answers questions like "which one?", "what kind?", "how many?". E.g., *happy, blue, three*.

4. Common Suffixes: 20

- Nouns: *-tion, -sion, -ment, -ness, -ity, -ance, -ence, -er, -or* (e.g., *information, development, happiness*). 21
- Verbs: *-ate, -ify, -ize, -en* (e.g., *activate, simplify, modernize*).
- Adjectives: *-ful, -less, -able, -ible, -ous, -ic, -al, -y* (e.g., *beautiful, hopeless, capable*).

• Key properties and identities: 22

- A word's function (noun, verb, adjective) can change based on context (e.g., "run" as a verb vs. "a run" as a noun). 23
- Understanding these parts of speech is fundamental to sentence construction and error identification.

• Common pitfalls: 24

- Confusing words that have similar forms but different parts of speech (e.g., "advice" (noun) vs. "advise" (verb)). 25
- Incorrectly identifying the function of a word in a sentence.

• Standard problem-solving techniques: 26

1. Identify the word's role in the sentence (e.g., is it naming something, performing an action, or describing something?). 27
2. Look for common suffixes that indicate a specific part of speech.
3. Test the word by replacing it with a clear example of a noun, verb, or adjective to see if the sentence structure

remains valid. 1

Passage Reading 2

Definition and core idea: This involves reading a given passage and answering comprehension questions based on its content, including main idea, specific details, inference, and author's tone. 3

• Key properties and identities: 4

- Questions can be direct (explicitly stated in the text) or inferential (requiring logical deduction from the text). 5
- Understanding the main idea and supporting details is crucial.

• Common pitfalls: 6

- Reading too slowly or too quickly without comprehension.
- Making assumptions not supported by the text.
- Getting bogged down by unfamiliar vocabulary.
- Misinterpreting the author's intent or tone.

• Standard problem-solving techniques: 7

1. **Skim the passage first:** Read quickly to get a general idea of the topic, main argument, and structure. Pay attention to the first and last sentences of paragraphs. 8
2. **Read the questions:** Understand what information you need to find. This helps you focus your reading.
3. **Scan for keywords:** For specific detail questions, scan the passage for keywords from the question.
4. **Read carefully for answers:** When you locate relevant sections, read them thoroughly to understand the context and find the exact answer.
5. **Inferential questions:** Look for clues, implications, and logical connections within the text. Avoid bringing in outside knowledge.
6. **Main idea questions:** Often found in the introduction or conclusion, or can be derived from summarizing the key points.
7. **Vocabulary in context:** Use surrounding words and sentences to deduce the meaning of unfamiliar words.

Phrasal Verb 9

Definition and core idea: A phrasal verb is a verb combined with a preposition or an adverb (or both) to create a new meaning that is often different from the original verb's meaning. 10

• Important formulas, theorems, and results: 11

1. *Verb + Preposition* (e.g., *look after* = care for) 12
2. *Verb + Adverb* (e.g., *break down* = stop functioning)
3. *Verb + Adverb + Preposition* (e.g., *look up to* = admire)

• Key properties and identities: 13

- Meanings are often idiomatic and cannot be deduced from the individual words.
- Can be separable (object can go between verb and particle) or inseparable.

• Common pitfalls: 14

- Assuming the literal meaning of the words.
- Confusing similar-looking phrasal verbs.
- Incorrectly using separable/inseparable phrasal verbs.

• Standard problem-solving techniques: 15

1. Memorize common phrasal verbs and their meanings. 16
2. Pay close attention to the context of the sentence to infer meaning.
3. If unsure, try to replace the phrasal verb with a single-word synonym to see if it fits.

Phrase Meaning 17

Definition and core idea: This involves understanding the meaning of idiomatic expressions, proverbs, or common phrases that often have a figurative meaning different from the literal interpretation of their individual words. 18

• Key properties and identities: 19

- Idioms are non-literal and culturally specific.
- Context is vital for interpreting the correct nuance of a phrase.

• Common pitfalls: 20

- Interpreting phrases literally.
- Confusing similar-sounding idioms.
- Not recognizing the figurative nature of the expression.

• Standard problem-solving techniques: 21

1. Read the sentence containing the phrase carefully to understand the overall context.
2. If the phrase is unfamiliar, try to infer its meaning from the surrounding words and the situation described.
3. Consider the options provided and see which one logically fits the context.
4. Eliminate options that represent a literal interpretation or are completely irrelevant.
5. Familiarity with common idioms and proverbs is the best preparation.

Prepositions 2

Definition and core idea: Prepositions are words that show the relationship between a noun or pronoun and other words in a sentence, indicating location, time, direction, or manner.

• Important formulas, theorems, and results: 4

1. Prepositions of Place: 5

- **in:** for enclosed spaces, cities, countries (e.g., *in the room, in India*).
- **on:** for surfaces, streets, specific days (e.g., *on the table, on Monday*).
- **at:** for specific points, addresses, events (e.g., *at the corner, at 5 PM*).

2. Prepositions of Time: 7

- **in:** for months, years, seasons, long periods (e.g., *in July, in 2023*).
- **on:** for specific days, dates (e.g., *on Friday, on August 15th*).
- **at:** for specific times (e.g., *at noon, at midnight*).

3. Prepositions of Direction: *to, into, onto, from, through, across*. E.g., *go to school, jump into the water*.

4. Fixed Prepositional Phrases: Many verbs, adjectives, and nouns are followed by specific prepositions (e.g., *afraid of, depend on, good at*).

• Key properties and identities: 11

- Prepositions always introduce a prepositional phrase, which includes a noun or pronoun (the object of the preposition).
- Their usage is often idiomatic and requires memorization.

• Common pitfalls: 13

- Confusing prepositions with similar meanings (e.g., "between" vs. "among").
- Incorrectly using fixed prepositional phrases.
- Omitting necessary prepositions or adding unnecessary ones.

• Standard problem-solving techniques: 15

1. Identify the relationship the preposition is trying to express (place, time, direction, etc.).
2. Check if the verb, adjective, or noun preceding the blank requires a specific preposition.
3. Consider the context and the object of the preposition.
4. Memorize common fixed prepositional phrases.

Pronouns 17

Definition and core idea: Pronouns are words that replace nouns to avoid repetition. They must agree with the nouns they refer to (antecedents) in number, gender, and case.

• Important formulas, theorems, and results: 19

1. Types of Pronouns: 20

- **Personal:** *I, you, he, she, it, we, they* (subjective); *me, you, him, her, it, us, them* (objective).
- **Possessive:** *mine, yours, his, hers, its, ours, theirs*.
- **Reflexive/Intensive:** *myself, yourself, himself, herself, itself, ourselves, yourselves, themselves*.
- **Relative:** *who, whom, whose, which, that*.
- **Demonstrative:** *this, that, these, those*.
- **Indefinite:** *everyone, somebody, nothing, all, many, few*.

2. Pronoun-Antecedent Agreement: A pronoun must agree with its antecedent in number (singular/plural) and gender (masculine/feminine/neuter). E.g., *The student submitted his assignment. The students submitted their assignments.*

3. Pronoun Case: 23

- **Subjective Case:** Used when the pronoun is the subject of a verb (e.g., *He went*).
- **Objective Case:** Used when the pronoun is the object of a verb or preposition (e.g., *She saw him. Give it to me*).

• Key properties and identities: 25

- Pronouns clarify sentences by preventing redundancy.
- Correct pronoun usage is essential for grammatical accuracy.

• Common pitfalls: 1

- Incorrect agreement in number or gender with the antecedent.
- Confusing subjective and objective cases (e.g., "between you and I" instead of "between you and me").
- Ambiguous pronoun reference (unclear what noun the pronoun refers to).
- Using "its" (possessive) vs. "it's" (it is).

• Standard problem-solving techniques: 3

1. Identify the pronoun and its antecedent (the noun it replaces).
2. Check for agreement in number (singular/plural) and gender.
3. Determine if the pronoun is acting as a subject or an object to choose the correct case.
4. Ensure the pronoun's reference is clear and unambiguous.

Sentence Ordering 5

Definition and core idea: Similar to Narrative Sequencing, this involves arranging jumbled parts of a single sentence (or a short paragraph) into a grammatically correct and logically coherent order.

• Key properties and identities: 7

- Focuses on sentence structure, grammatical rules, and logical flow within a single sentence or a very short set of related sentences.
- Often involves identifying the subject, verb, and object, and then arranging modifiers correctly.

• Common pitfalls: 9

- Creating grammatically incorrect structures.
- Producing a sentence that is grammatically correct but makes no logical sense.
- Misplacing modifiers, leading to ambiguity.

• Standard problem-solving techniques: 11

1. **Identify the subject and main verb:** These usually form the core of the sentence.
2. **Look for introductory phrases/clauses:** These often come at the beginning.
3. **Identify conjunctions:** Words like *and*, *but*, *or*, *because*, *while* connect clauses or phrases.
4. **Check for pronoun references:** Ensure pronouns follow their antecedents.
5. **Consider logical flow:** What event or idea logically precedes another?
6. **Read the assembled sentence:** Ensure it is grammatically correct, makes sense, and flows naturally.

Statement Sufficiency 13

Definition and core idea: In this type of question, a question is given, followed by two or more statements. You need to determine whether the information in the statements is sufficient to answer the question.

• Key properties and identities: 15

- Focus is on sufficiency, not necessarily on finding the answer itself.
- Statements are evaluated independently first, then together if neither is sufficient alone.

• Common pitfalls: 17

- Trying to solve the question instead of just checking sufficiency.
- Assuming information not explicitly given in the statements.
- Incorrectly combining statements.

• Standard problem-solving techniques: 18

1. **Analyze the question:** Understand exactly what information is required to answer it.
2. **Evaluate Statement 1 alone:** Can the question be answered using only the information in Statement 1? Ignore Statement 2.
3. **Evaluate Statement 2 alone:** Can the question be answered using only the information in Statement 2? Ignore Statement 1.
4. **Evaluate Statements 1 and 2 together:** If neither statement alone is sufficient, check if combining them provides enough information.

5. Select the appropriate option: 20

- Statement 1 alone is sufficient.
- Statement 2 alone is sufficient.
- Either Statement 1 alone or Statement 2 alone is sufficient.
- Both Statement 1 and Statement 2 together are sufficient (but neither alone).
- Neither Statement 1 nor Statement 2 nor both together are sufficient.

Statements Follow 22

Definition and core idea: This involves analyzing a set of given statements (premises) and determining which

conclusion(s) logically follow from them, often based on principles of deductive reasoning. 1

• **Important formulas, theorems, and results:** 2

1. **Deductive Reasoning:** If the premises are true, the conclusion must be true. 3

- Example: *All A are B. All B are C. Therefore, All A are C.*

2. **Categorical Syllogisms:** Often involve "All," "No," "Some" statements. 4

- *All X are Y* (Universal Affirmative)
- *No X are Y* (Universal Negative)
- *Some X are Y* (Particular Affirmative)
- *Some X are not Y* (Particular Negative)

3. **Venn Diagrams:** A useful tool for visually representing relationships between categories and testing the validity of conclusions. 5

• **Key properties and identities:** 6

- Conclusions must be derived solely from the given statements, without external knowledge.
- Focus on logical necessity, not probability or common sense.

• **Common pitfalls:** 7

- Bringing in outside knowledge or personal opinions. 8
- Making assumptions not supported by the statements.
- Confusing "some" with "all" or "none."
- Errors in interpreting negative statements.

• **Standard problem-solving techniques:** 9

1. **Read statements carefully:** Understand the exact meaning of each premise. 10
2. **Use Venn Diagrams:** Draw circles to represent the categories and their relationships as described by the statements.
3. **Test each conclusion:** Check if the conclusion is necessarily true based on your diagram or logical deduction. If there's even one scenario where the conclusion is false, it does not follow.
4. **Avoid assumptions:** Stick strictly to the information provided in the statements. 11

Synonyms 12

Definition and core idea: Synonyms are words that have the same or very similar meanings. Understanding synonyms expands vocabulary and allows for varied expression. 13

• **Key properties and identities:** 14

- Few words are perfect synonyms; most have slightly different connotations or contexts of use. 15
- Can often be identified by root words, prefixes, and suffixes.

• **Common pitfalls:** 16

- Choosing a word that is related but not a true synonym.
- Ignoring the context in which the word is used.
- Misinterpreting the nuance of the given word.

• **Standard problem-solving techniques:** 17

1. Understand the precise meaning of the given word. 18
2. Consider the context if the word has multiple meanings.
3. Look for words with similar roots or derivations.
4. Substitute each option into a sentence where the original word is used to see which one fits best.
5. Eliminate options that are antonyms or unrelated.

Tenses 19

Definition and core idea: Tenses indicate the time an action occurs (past, present, future) and its aspect (simple, continuous, perfect, perfect continuous), showing whether an action is completed, ongoing, or habitual. 20

• **Important formulas, theorems, and results:** 21

1. **Simple Present:** $S + V1 + (s/es)$ (for 3rd person singular). Used for habits, facts, scheduled events. E.g., *He walks.* 22
2. **Present Continuous:** $S + is/am/are + V - ing$. Used for actions happening now or temporary actions. E.g., *He is walking.*
3. **Present Perfect:** $S + has/have + V3$. Used for actions completed in the past but with present relevance. E.g., *He has walked.*
4. **Present Perfect Continuous:** $S + has/have + been + V - ing$. Used for actions that started in the past and continue to the present. E.g., *He has been walking.*

5. **Simple Past:** $S + V2$. Used for completed actions in the past. E.g., *He walked*. 1
6. **Past Continuous:** $S + was/were + V - ing$. Used for actions ongoing in the past. E.g., *He was walking*. 2
7. **Past Perfect:** $S + had + V3$. Used for an action completed before another past action. E.g., *He had walked* before I arrived. 3
8. **Past Perfect Continuous:** $S + had + been + V - ing$. Used for an action ongoing up to a point in the past. E.g., *He had been walking* for an hour when I saw him. 3
9. **Simple Future:** $S + will/shall + V1$. Used for future actions. E.g., *He will walk*. 4
10. **Future Continuous:** $S + will/shall + be + V - ing$. Used for actions ongoing at a specific time in the future. E.g., *He will be walking*. 5
11. **Future Perfect:** $S + will/shall + have + V3$. Used for actions that will be completed by a certain time in the future. E.g., *He will have walked* 10 km by then. 6
12. **Future Perfect Continuous:** $S + will/shall + have + been + V - ing$. Used for actions that will have been ongoing up to a point in the future. E.g., *He will have been walking* for two hours. 7

• **Key properties and identities:** 8

- Tense consistency is crucial within a sentence or paragraph. 9
- Time markers (e.g., yesterday, now, next week, since, for) often indicate the appropriate tense.

• **Common pitfalls:**

- Incorrectly mixing tenses within a sentence. 10
- Confusing similar tenses (e.g., simple past vs. present perfect).
- Incorrect verb forms (V1, V2, V3).

• **Standard problem-solving techniques:** 11

1. Identify the time frame of the action (past, present, future). 12
2. Determine the aspect (simple, continuous, perfect, perfect continuous) based on whether the action is completed, ongoing, or has relevance to another time.
3. Look for time expressions or other verbs in the sentence that provide clues about the correct tense.
4. Ensure tense consistency, especially in complex sentences or reported speech.

Verbal Reasoning 13

Definition and core idea: Verbal Reasoning assesses the ability to understand and logically process information presented in words, including analogies, critical reasoning, and logical deduction from textual information. 14

• **Key properties and identities:** 15

- Requires analytical thinking and the ability to identify patterns, relationships, and logical fallacies. 16
- Often involves drawing conclusions, identifying assumptions, or evaluating arguments.

• **Common pitfalls:** 17

- Making assumptions not explicitly stated. 18
- Confusing correlation with causation.
- Allowing personal bias to influence judgment.
- Misinterpreting logical connectors (e.g., "if...then," "unless").

• **Standard problem-solving techniques:** 19

1. **Read carefully:** Understand the exact meaning of all statements and the question. 20
2. **Identify the core argument/relationship:** For analogies, determine the relationship between the first pair. For critical reasoning, identify the premise(s) and conclusion.
3. **Break down complex statements:** Simplify sentences to their core logical components.
4. **Look for keywords:** Words like *therefore*, *because*, *however*, *if*, *then*, *unless*, *all*, *some*, *none* are crucial for logical deduction.
5. **Test options:** For multiple-choice questions, evaluate each option against the given information.
6. **Avoid outside knowledge:** Base your reasoning strictly on the provided text.

Word Meaning 21

Definition and core idea: This involves understanding the precise definition of a word, often in a given context, and choosing the option that best reflects that meaning. 22

• **Key properties and identities:** 23

- Words can have multiple meanings; context determines the relevant one. 24
- Knowledge of root words, prefixes, and suffixes can help deduce meanings.

• **Common pitfalls:**

- Choosing a common meaning of a word when the context implies a less common one. 25
- Misinterpreting the nuance or connotation of a word.

- Ignoring contextual clues. 1

• Standard problem-solving techniques: 2

1. Read the sentence or phrase containing the word carefully.
2. Try to infer the meaning of the word from its context (surrounding words, overall sentence meaning).
3. Look for synonyms or antonyms within the sentence that might provide clues.
4. Break down the word into its root, prefix, and suffix if applicable.
5. Substitute each option into the sentence to see which one makes the most sense.
6. Eliminate options that are clearly incorrect or irrelevant.

3

Word Pairs 4

Definition and core idea: This question type presents a pair of words that share a specific relationship, and you must identify another pair of words from the options that shares the same relationship (analogies).

• Important formulas, theorems, and results: 6

1. Common Relationship Types:

- **Synonym:** *Happy : Joyful*
- **Antonym:** *Hot : Cold*
- **Part-Whole:** *Finger : Hand*
- **Cause-Effect:** *Rain : Flood*
- **Tool-User:** *Pen : Writer*
- **Object-Function:** *Knife : Cut*
- **Category-Member:** *Fruit : Apple*
- **Intensity:** *Warm : Hot*
- **Producer-Product:** *Baker : Bread*
- **Action-Object:** *Read : Book*

• Key properties and identities: 7

- The relationship must be consistent across both pairs.
- The order of words in the pair often matters (e.g., cause-effect is different from effect-cause).

8

• Common pitfalls: 9

- Identifying a weak or incorrect relationship in the original pair.
- Finding a relationship in the options that is similar but not identical to the original.
- Ignoring the order of the words in the pair.

10

• Standard problem-solving techniques: 11

1. **Identify the precise relationship:** Clearly define how the two words in the given pair are related. Be specific (e.g., "is a type of," "is used to," "is the opposite of").
2. **Formulate a sentence:** Create a simple sentence that expresses the relationship (e.g., "A finger is a part of a hand").
3. **Apply the sentence to options:** Test each option by plugging its words into your sentence. The option that fits best is likely the answer.
4. **Consider nuances:** If multiple options seem plausible, re-evaluate the strength and specificity of the relationship.
5. **Check for order:** Ensure the order of the words in the option pair matches the order of the relationship in the original pair.

12

Quick Formula Reference 13

For Verbal Aptitude, "formulas" refer to core rules, structures, and patterns. Here's a consolidated list: 14

• Articles: 15

- 'a': Singular, countable, consonant sound.
- 'an': Singular, countable, vowel sound.
- 'the': Specific, unique, or previously mentioned.
- Zero Article: General plural/uncountable/abstract nouns, proper nouns.

16

• Comparative Forms:

- 1-syllable: *Adj + -er / -est*
- 2-syllable ending in -y: *Adj(-y → -i) + -er / -est*
- 2+ syllables: *more / most + Adj*
- Irregular: *good → better → best, bad → worse → worst*, etc.

17

• English Grammar (Core):

- Sentence Structure: *S + V + O* (common).

18

- Subject-Verb Agreement: *SingularS* → *SingularV*, *PluralS* → *PluralV*. 1
- Tense Consistency: Maintain consistent tense.
- Parallelism: Similar items in same grammatical form.

• Pronouns: 2

- Agreement: Pronoun matches antecedent in number, gender.
- Case: Subjective (*I, he, she, we, they*) for subjects; Objective (*me, him, her, us, them*) for objects. 3

• Tenses (Basic Structures): 4

- Simple Present: $S + V1 + (s/es)$
- Present Continuous: $S + is/am/are + V - ing$
- Present Perfect: $S + has/have + V3$
- Simple Past: $S + V2$
- Past Continuous: $S + was/were + V - ing$
- Past Perfect: $S + had + V3$
- Simple Future: $S + will/shall + V1$

• Phrasal Verbs: *Verb + Preposition/Adverb* (often idiomatic meaning). 5

• Statement Sufficiency: Evaluate Statement 1 alone, then Statement 2 alone, then both together.

• Statements Follow (Deductive Logic): If $Premise_1$ and $Premise_2$ are true, then *Conclusion* MUST be true. Use Venn Diagrams.

• Word Pairs (Analogies): Identify the specific relationship $A : B$ and find $C : D$ with the same relationship.

Important Tips for GATE 6

1. **Build a Strong Vocabulary:** Many questions (Synonyms, Antonyms, Word Meaning, Most Appropriate Word, Word Pairs, Phrasal Verbs, Phrase Meaning) directly test vocabulary. Read widely, use flashcards, and learn root words, prefixes, and suffixes. 7
2. **Master Grammar Rules:** Questions on Articles, Comparative Forms, English Grammar, Grammatical Error, Incorrect Sentence Part, Prepositions, Pronouns, and Tenses require a solid understanding of fundamental grammar. Practice identifying common errors. 8
3. **Read Actively and Critically:** For Passage Reading, Narrative Sequencing, and Verbal Reasoning, don't just read passively. Actively look for main ideas, supporting details, logical connections, transition words, and author's tone. Practice skimming and scanning. 9
4. **Context is King:** For vocabulary-based questions and even grammar, the context of the sentence or passage is crucial. A word's meaning or the correct grammatical form can change depending on its usage. Always read the full sentence. 10
5. **Practice Time Management:** Verbal Aptitude questions can be time-consuming, especially Passage Reading and Sentence Ordering. Practice solving questions under timed conditions to improve speed and accuracy. Don't get stuck on one question; move on and return if time permits. 11
6. **Eliminate Options Strategically:** For multiple-choice questions, if you're unsure of the answer, try to eliminate obviously incorrect options first. This increases your chances of selecting the correct answer, especially if negative marking is less severe. 12
7. **Don't Overthink Verbal Reasoning:** For Statement Sufficiency and Statements Follow, stick strictly to the information given in the statements. Do not bring in outside knowledge or make assumptions. Focus purely on logical deduction. 13
8. **Review Common Pitfalls:** Be aware of common errors like subject-verb agreement, pronoun-antecedent agreement, tense shifts, double comparatives, and literal interpretation of idioms. Regularly review these tricky areas. 14

11.1

Antonyms (7)

15

 **Practice Test:** Test 1 (7Q) 16

11.1.1 Antonyms: GATE CSE 2011 | Question: 60



Choose the word from the options given below that is most nearly opposite in the meaning to the given word

Amalgamate

- A. merge B. split C. collect D. separate

gatecse-2011 verbal-aptitude antonyms normal

Answer key 

11.1.2 Antonyms: GATE CSE 2014 | Set 3 | GA | Question: 3



Choose the word that is opposite in meaning to the word “coherent”. 1

- A. sticky B. well-connected C. rambling D. friendly 2

gatecse-2014-set3 verbal-aptitude antonyms easy 3

Answer key

11.1.3 Antonyms: GATE CSE 2026 | Set 1 | GA | Question: 1



The antonym of the word protagonist is _____. 4

- A. agnostic B. antagonist C. arsonist D. anarchist 5

gatecse-2026-set1 verbal-aptitude antonyms easy one-mark 6

Answer key 7

11.1.4 Antonyms: GATE DA 2026 | GA | Question: 1



Verbosity : Brevity :: Insolence : _____ 8

Choose the word that best fills the blank.

- A. Innocence B. Respect C. Solace D. Wealth 9

gateda-2026 verbal-aptitude antonyms one-mark 10

Answer key

11.1.5 Antonyms: GATE2011 AG: GA-2



Choose the word from the questions given below that is most nearly opposite in meaning to the given word: 11

Frequency

- A. periodicity B. rarity C. gradualness D. persistency 13

general-aptitude verbal-aptitude gate2011-ag antonyms 14

Answer key

11.1.6 Antonyms: GATE2011 GG: GA-2

15



Choose the word from the options given below that is most nearly opposite in meaning to the given word: 16

Polemical 17

- A. imitative B. conciliatory C. truthful D. ideological 18

gate2011-gg verbal-aptitude antonyms

Answer key

11.1.7 Antonyms: GATE2011 MN: GA-56

19



Choose the word from the options given below that is most nearly opposite in meaning to the given word: 20

Deference

- A. aversion B. resignation C. suspicion D. contempt 21

verbal-aptitude gate2011-mn antonyms 22

Answer key

11.2

Articles (2)

11.2.1 Articles: GATE CSE 2016 | Set 2 | GA | Question: 01



The man who is now Municipal Commissioner worked as _____.

- A. the security guard at a university
C. a security guard at university
- B. a security guard at the university
D. the security guard at the university

gatecse-2016-set2 verbal-aptitude english-grammar normal articles 2

Answer key

11.2.2 Articles: GATE CSE 2025 | Set 1 | GA | Question: 1



Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man.

Select the option with the correct sequence of articles to fill in the blanks. 4

- A. a; a; an
C. a; an; a
- B. the; an; a
D. an; an; a

gatecse2025-set1 verbal-aptitude articles easy one-mark 6

Answer key 7

11.3 Comparative Forms (2)

Practice Test: Test 1 (7Q)

8

11.3.1 Comparative Forms: GATE CSE 2015 | Set 1 | GA | Question: 1



Didn't you buy _____ when you went shopping? 10

- A. any paper
B. much paper
C. no paper
D. a few paper

gatecse-2015-set1 verbal-aptitude easy english-grammar comparative-forms 12

Answer key

11.3.2 Comparative Forms: GATE CSE 2021 | Set 2 | GA | Question: 1



Gauri said that she can play the keyboard _____ her sister. 13

- A. as well as
B. as better as
C. as nicest as
D. as worse as

gatecse-2021-set2 verbal-aptitude comparative-forms one-mark 15

Answer key 16

11.4 English Grammar (5)

Practice Tests: Test 1 (15Q) Test 2 (11Q) 17

11.4.1 English Grammar: GATE CSE 2015 | Set 2 | GA | Question: 1



We _____ our friends's birthday and we _____ how to make it up to him. 19

- A. completely forgot --- don't just know
C. completely forgot --- just don't know
- B. forgot completely --- don't just know
D. forgot completely --- just don't know

gatecse-2015-set2 verbal-aptitude easy english-grammar 21

Answer key

11.4.2 English Grammar: GATE CSE 2015 | Set 3 | GA | Question: 2



The Tamil version of _____ John Abraham-starrer *Madras Cafe* _____ cleared by the Censor Board with no cuts last week, but the film's distributor _____ no takers among the exhibitors for a release in Tamilnadu _____ this Friday.

- A. Mr., was, found, on
C. the, was, found, on
- B. a, was, found, at
D. a, being, find, at

gatecse-2015-set3 verbal-aptitude normal english-grammar

Answer key

11.4.3 English Grammar: GATE CSE 2017 | Set 2 | GA | Question: 2



Saturn is _____ to be seen on a clear night with the naked eye. **1**

- A. enough bright B. bright enough C. as enough bright D. bright as enough **2**

gatecse-2017-set2 verbal-aptitude english-grammar

Answer key

11.4.4 English Grammar: GATE CSE 2020 | GA | Question: 1



Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks **3**

- A. during, for B. for, since C. for, in D. within, for **4**

gatecse-2020 verbal-aptitude english-grammar one-mark **5**

Answer key

11.4.5 English Grammar: GATE CSE 2021 | Set 1 | GA | Question: 3



Consider the following sentences: **6**

- i. Everybody in the class is prepared for the exam. **7**
ii. Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the **CORRECT** observation about the above two sentences? **8**

- A. (i) is grammatically correct and (ii) is unambiguous **9**
B. (i) is grammatically incorrect and (ii) is unambiguous
C. (i) is grammatically correct and (ii) is ambiguous
D. (i) is grammatically incorrect and (ii) is ambiguous

gatecse-2021-set1 verbal-aptitude english-grammar one-mark

Answer key

11.5

Grammatical Error (6)

Practice Test: Test 1 (11Q) **10**

11.5.1 Grammatical Error: GATE CSE 2012 | Question: 59 **11**



Choose the grammatically **INCORRECT** sentence: **12**

- A. They gave us the money back less the service charges of Three Hundred rupees. **13**
B. This country's expenditure is not less than that of Bangladesh.
C. The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.
D. This country's expenditure on educational reforms is very less.

gatecse-2012 verbal-aptitude grammatical-error normal

Answer key

11.5.2 Grammatical Error: GATE CSE 2013 | Question: 60



Choose the grammatically **INCORRECT** sentence:

- A. He is of Asian origin. B. They belonged to Africa.
C. She is an European. D. They migrated from India to Australia.

gatecse-2013 verbal-aptitude grammatical-error normal

Answer key

11.5.3 Grammatical Error: GATE CSE 2015 | Set 2 | GA | Question: 10



Out of the following 4 sentences, select the most suitable sentence with respect to grammar and usage: 1

- A. Since the report lacked needed information, it was of no use to them. 2
- B. The report was useless to them because there were no needed information in it.
- C. Since the report did not contain the needed information, it was not real useful to them.
- D. Since the report lacked needed information, it would not had been useful to them.

gatecse-2015-set2 verbal-aptitude normal english-grammar grammatical-error 3

Answer key

11.5.4 Grammatical Error: GATE CSE 2016 | Set 1 | GA | Question: 01



Out of the following four sentences, select the most suitable sentence with respect to grammar and usage. 5

- A. I will not leave the place until the minister does not meet me. 6
- B. I will not leave the place until the minister doesn't meet me.
- C. I will not leave the place until the minister meet me.
- D. I will not leave the place until the minister meets me.

gatecse-2016-set1 verbal-aptitude english-grammar easy grammatical-error 7

Answer key

11.5.5 Grammatical Error: GATE2012 AR: GA-3



Choose the grammatically CORRECT sentence: 8

- A. He laid in bed till 8 o'clock in the morning. 9
- B. He layed in bed till 8 o'clock in the morning.
- C. He lain in bed till 8 o'clock in the morning.
- D. He lay in bed till 8 o'clock in the morning.

gate2012-ar verbal-aptitude english-grammar easy grammatical-error 10

Answer key

11.5.6 Grammatical Error: GATE2013 EE: GA-4



Choose the grammatically CORRECT sentence: 11

- A. Two and two add four. 12
- B. Two and two become four.
- C. Two and two are four.
- D. Two and two make four.

gate2013-ee english-grammar verbal-aptitude grammatical-error 13

Answer key 14

11.6

Incorrect Sentence Part (5)

Practice Test: Test 1 (6Q)

15

11.6.1 Incorrect Sentence Part: GATE CSE 2014 | Set 3 | GA | Question: 1



16

While trying to collect I an envelope from under the table, Mr. X fell down and was losing consciousness. 17

IV

Which one of the above underlined parts of the sentence is NOT appropriate?

- A. I 18
- B. II
- C. III
- D. IV

gatecse-2014-set3 verbal-aptitude easy english-grammar incorrect-sentence-part 19

Answer key

11.6.2 Incorrect Sentence Part: GATE2012 AR: GA-4



Which one of the parts (A, B, C, D) in the sentence contains an **ERROR**? **1**

No sooner had the doctor seen the results of the blood test, than he suggested the patient to see the specialist. **2**

- A. no sooner had
C. suggested the patient
B. results of the blood test
D. see the specialist **3**

gate2012-ar verbal-aptitude incorrect-sentence-part **4**

Answer key

11.6.3 Incorrect Sentence Part: GATE2012 CY: GA-2



One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT**? **5**

I requested that he should be given the driving test today instead of tomorrow. **6**

- A. requested that
C. the driving test
B. should be given
D. instead of tomorrow **7**

gate2012-cy verbal-aptitude incorrect-sentence-part **8**

Answer key

11.6.4 Incorrect Sentence Part: GATE2013 AE: GA-4



All engineering students should learn mechanics, mathematics and how to do computation. **9**

I II III IV

Which of the above underlined parts of the sentence is not appropriate? **10**

- a. I b. II c. III d. IV **11**

gate2013-ae verbal-aptitude incorrect-sentence-part

Answer key

11.6.5 Incorrect Sentence Part: GATE2013 CE: GA-2



The professor ordered to the students to go out of the class. **12**

I II III IV

Which of the above underlined parts of the sentence is grammatically incorrect?

- A. I B. II C. III D. IV **13**

gate2013-ce incorrect-sentence-part verbal-aptitude **14**

Answer key **15**

11.7

Most Appropriate Word (47)

Practice Tests: Test 1 (15Q) Test 10 (3Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (15Q) Test 6 (15Q) Test 7 (15Q) **16**
Test 8 (15Q) Test 9 (15Q)

11.7.1 Most Appropriate Word: GATE CSE 2010 | Question: 56 **17**



Choose the most appropriate word from the options given below to complete the following sentence: **18**

His rather casual remarks on politics _____ his lack of seriousness about the subject.

- A. masked B. belied C. betrayed D. suppressed **19**

gatecse-2010 verbal-aptitude most-appropriate-word normal **20**

Answer key

11.7.2 Most Appropriate Word: GATE CSE 2010 | Question: 58



Choose the most appropriate word from the options given below to complete the following sentence:

If we manage to _____ our natural resources, we would leave a better planet for our children.

A. uphold B. restrain C. cherish D. conserve 1

gatecse-2010 verbal-aptitude most-appropriate-word easy

Answer key

11.7.3 Most Appropriate Word: GATE CSE 2011 | Question: 58

Choose the most appropriate word(s) from the options given below to complete the following sentence. 2

I contemplated _____ Singapore for my vacation but decided against it.

A. to visit B. having to visit C. visiting D. for a visit 3

gatecse-2011 verbal-aptitude most-appropriate-word easy 4

Answer key

11.7.4 Most Appropriate Word: GATE CSE 2011 | Question: 59

Choose the most appropriate word from the options given below to complete the following sentence.

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.

A. hyperbolic B. restrained C. argumentative D. indifferent 7

gatecse-2011 verbal-aptitude most-appropriate-word normal 8

Answer key

11.7.5 Most Appropriate Word: GATE CSE 2012 | Question: 57

Choose the most appropriate alternative from the options given below to complete the following sentence: 10

Despite several _____ the mission succeeded in its attempt to resolve the conflict.

A. attempts B. setbacks C. meetings D. delegations 11

gatecse-2012 verbal-aptitude easy most-appropriate-word 12

Answer key

11.7.6 Most Appropriate Word: GATE CSE 2012 | Question: 60

Choose the most appropriate alternative from the options given below to complete the following sentence: 14

Suresh's dog is the one _____ was hurt in the stampede.

A. that B. which C. who D. whom 15

gatecse-2012 verbal-aptitude most-appropriate-word normal 16

Answer key

11.7.7 Most Appropriate Word: GATE CSE 2014 | Set 1 | GA | Question: 2

Choose the most appropriate word from the options given below to complete the following sentence. 18

She could not understand the judges awarding her the first prize, because she thought that her performance was quite _____ 19

A. superb B. medium C. mediocre D. exhilarating 20

gatecse-2014-set1 verbal-aptitude most-appropriate-word easy 21

Answer key

11.7.8 Most Appropriate Word: GATE CSE 2014 | Set 2 | GA | Question: 1

Choose the most appropriate phrase from the options given below to complete the following sentence. 22

India is a post-colonial country because

- A. it was a former British colony
B. Indian Information Technology professionals have colonized the world
C. India does not follow any colonial practices
D. India has helped other countries gain freedom

1

gatecse-2014-set2 verbal-aptitude most-appropriate-word easy

Answer key

11.7.9 Most Appropriate Word: GATE CSE 2015 | Set 2 | GA | Question: 4

A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

- A. fabric B. textile C. fiber D. apparel

gatecse-2015-set2 verbal-aptitude easy most-appropriate-word

Answer key

11.7.10 Most Appropriate Word: GATE CSE 2015 | Set 3 | GA | Question: 3

Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.

- A. related B. extraneous C. outside D. useful

gatecse-2015-set3 verbal-aptitude normal most-appropriate-word

Answer key

11.7.11 Most Appropriate Word: GATE CSE 2016 | Set 1 | GA | Question: 03

Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

- A. figurative B. collateral C. literal D. figurine

gatecse-2016-set1 verbal-aptitude normal most-appropriate-word

Answer key

11.7.12 Most Appropriate Word: GATE CSE 2017 | Set 1 | GA | Question: 2

Research in the workplace reveals that people work for many reasons _____.

- A. money beside B. beside money
C. money besides D. besides money

gatecse-2017-set1 general-aptitude verbal-aptitude english-grammar most-appropriate-word

Answer key

11.7.13 Most Appropriate Word: GATE CSE 2018 | GA | Question: 1

"From where are they bringing their books? _____ bringing _____ books from _____"

The words that best fill the blanks in the above sentence are

- A. Their, they're, there B. They're, their, there
C. There, their, they're D. They're, there, there

gatecse-2018 verbal-aptitude most-appropriate-word easy one-mark

Answer key

11.7.14 Most Appropriate Word: GATE CSE 2018 | GA | Question: 2

A _____ investigation can sometimes yield new facts, but typically organized ones are more successful.

The word that best fills the blank in the above sentence is

- A. meandering B. timely C. consistent D. systematic

Answer key

11.7.15 Most Appropriate Word: GATE CSE 2019 | GA | Question: 1

The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

- A. break down B. break C. breaks down D. breaks 2

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark 3

Answer key

11.7.16 Most Appropriate Word: GATE CSE 2019 | GA | Question: 2

The search engine's business model _____ around the fulcrum of trust. 5

- A. revolves B. plays C. sinks D. bursts 6

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark 7

Answer key 8

11.7.17 Most Appropriate Word: GATE CSE 2019 | GA | Question: 5

A court is to a judge as _____ is to a teacher 10

- A. a student B. a punishment C. a syllabus D. a school 11

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark 12

Answer key 13

11.7.18 Most Appropriate Word: GATE CSE 2022 | GA | Question: 1

The _____ is too high for it to be considered _____. 15

- A. fair / fare B. faer / fair C. fare / fare D. fare / fair 16

gatecse-2022 verbal-aptitude most-appropriate-word one-mark 17

Answer key

11.7.19 Most Appropriate Word: GATE CSE 2023 | GA | Question: 1

We reached the station late, and _____ missed the train. 19

- A. near B. nearly C. utterly D. mostly 20

gatecse-2023 verbal-aptitude most-appropriate-word one-mark 21

Answer key

11.7.20 Most Appropriate Word: GATE CSE 2024 | Set 1 | GA | Question: 1

If ' → ' denotes increasing order of intensity, then the meaning of the words 23

[dry → arid → parched] is analogous to [diet → fast → _____].

Which one of the given options is appropriate to fill the blank?

- A. starve B. reject C. feast D. deny 24

gatecse-2024-set1 verbal-aptitude most-appropriate-word one-mark 25

Answer key

11.7.21 Most Appropriate Word: GATE CSE 2024 | Set 1 | GA | Question: 6

In the given text, the blanks are numbered (i) - (iv). Select the best match for all the blanks.

Steve was advised to keep his head (i) before heading (ii) to bat; for, while he had a head (iii) batting, he could only do so with a cool head (iv) his shoulders.

- A. (i) down (ii) down (iii) on (iv) for
C. (i) down (ii) out (iii) for (iv) on
- B. (i) on (ii) down (iii) for (iv) on
D. (i) on (ii) out (iii) on (iv) for

gatecse-2024-set1 verbal-aptitude most-appropriate-word two-marks 2

Answer key

11.7.22 Most Appropriate Word: GATE CSE 2024 | Set 2 | GA | Question: 1



If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted].

Which one of the given options is appropriate to fill the blank? 4

- A. phased B. phrased C. fazed D. fused 5

gatecse-2024-set2 verbal-aptitude most-appropriate-word one-mark 6

Answer key

11.7.23 Most Appropriate Word: GATE CSE 2025 | Set 1 | GA | Question: 2



The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains.

Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence. 9

- A. visionary B. shortsighted C. progressive D. innovative 10

gatecse2025-set1 verbal-aptitude most-appropriate-word easy one-mark 11

Answer key

11.7.24 Most Appropriate Word: GATE CSE 2025 | Set 2 | GA | Question: 1



Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered.

Select the most appropriate option to complete the above sentence.

- A. ambivalence B. satisfaction C. resolve D. revolve 13

gatecse2025-set2 verbal-aptitude most-appropriate-word easy one-mark 14

Answer key

11.7.25 Most Appropriate Word: GATE DA 2025 | GA | Question: 1



Courage : Bravery :: Yearning : _____ 16

Select the most appropriate option to complete the analogy.

- A. Longing B. Yelling C. Yawning D. Glaring 17

gateda-2025 verbal-aptitude most-appropriate-word easy one-mark 18

Answer key 19

11.7.26 Most Appropriate Word: GATE DS&AI 2024 | GA | Question: 1



If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft].

Which one of the given options is appropriate to fill the blank? 22

- A. frown B. fawn C. vein D. vain 23

gate-ds-ai-2024 verbal-aptitude most-appropriate-word one-mark

Answer key